

Optimization and Performance Analysis of Resource Allocation in Quantum-Centric Supercomputing Environments

Squillante and Tantawi and potentially others

January 2026

Abstract

Advances in quantum computing hardware and quantum algorithms are likely to cause major paradigm shifts in high-performance supercomputing environments. This shift includes foundational changes to core designs for high-performance systems that integrate both quantum and classical computational substrates via a combination of quantum processing units, graphical processing units, and central processing units. This shift also includes foundational changes to core designs for high-performance application workloads that combine different patterns of computation across various quantum and classical processing units, resulting in jobs that require only classical resources, only quantum resources, or a combination of both. While the optimization and performance analysis of resource allocation in high-performance supercomputing environments under classical-only workloads may have received considerable attention, our understanding of these matters is far more limited and largely unexplored in quantum-centric supercomputing environments under quantum-only and classical-quantum-combined application workloads, especially the latter.

In this paper, we derive a general theoretical foundation for the mathematical optimization and performance analysis of resource allocation in quantum-centric supercomputing environments. Our theoretical foundation includes mathematical models of application workload patterns, together with a mathematical analysis of basic computational phases at the core of these patterns. We derive a mathematical analysis of the performance of quantum-centric supercomputing environments under a reservation-based resource allocation policy, which is not common in practice. We then formulate a general resource capacity allocation optimization problem in the context of quantum-centric supercomputing environments, from which we derive an efficient numerical solution for the optimal policy that differs significantly from the reservation-based resource allocation policy. Numerical experiments illustrate the performance of the core computational phases across different application workloads, further demonstrating the strong accuracy of our closed-form approximations for the corresponding fork-join queueing networks, which are of interest in their own right. Numerical experiments also illustrate the optimal resource capacity allocation and associated performance across different

application workloads, including performance comparisons with the reservation-based resource allocation policy under the same resource capacity allocations.

1 Introduction

The evolution and growth of quantum computing has been tremendous since its origins, starting with the original proposal by Benioff in 1980 for a quantum version of a Turing machine based on the equations of quantum mechanics [9]. This was shortly followed by Feynman’s keynote at the First Conference on the Physics of Computation, jointly organized in 1981 by IBM and MIT, in which he proposed a “quantum simulator” that, unlike a universal Turing machine, specializes in simulating quantum mechanical phenomena [15]. Feynman closed this keynote with his now-famous remark: “. . . nature isn’t classical, dammit, and if you want to make a simulation of nature, you’d better make it quantum mechanical, and by golly it’s a wonderful problem, because it doesn’t look so easy.”

Since then, quantum computing has increasingly captured broad attention across various fields of science, engineering, and business due to its potential to solve certain problems much faster than classical computers. Over the past few years, there have been immense advances in quantum computing hardware. This includes quantum computers that use logical quantum gates to execute algorithms, such as those based on superconducting quantum bits (e.g., IBM, Google, Rigetti), trapped-ion systems (e.g., IonQ, Quantinuum), cold-atom devices (e.g., QuEra), and photonic devices (e.g., Xanadu, PsiQuantum). These quantum computers are capable in principle of executing any algorithm in the complexity class for quantum polynomial-time computation, known as BQP.

In combination with such progress and growth in quantum computing hardware, there have been fundamental advances in quantum algorithms across different computational problem domains which offer tremendous potential for truly significant performance advantages in the execution of such algorithms on quantum computers over their classical counterparts for certain problems. This includes examples in factoring large numbers [57,58], unstructured search [20,21], parameter estimation [10], linear systems of equations [23], Monte Carlo methods [30,38], finance [48,65], Bayesian experimental design [19], and optimization [1,24]. This even includes recent examples in which quantum computers have demonstrated benefit over brute-force classical approaches in simulating quantum systems [28,52]. These events and trends have helped to bring about major paradigm shifts in the evolution of high-performance supercomputing environments that integrates both quantum computing and classical high-performance computing through a combination of quantum processing units (QPUs), graphical processing units (GPUs), and central processing units (CPUs) [8,36,54,56,59,62]. Such paradigm shifts include foundational changes in core designs related to high-performance computing system infrastructures. In fact, Seelam et al. [54] have even defined the first reference architecture for so-called quantum-centric supercomputing environments. These paradigm shifts also include foundational changes in core designs related to high-performance application workloads that combine different patterns of computation across the various QPUs, GPUs, and CPUs. This results in application workloads comprising jobs that require

only classical GPUs and CPUs, jobs that require only quantum QPUs, or jobs that require a mixture of classical GPUs and CPUs and quantum QPUs. At the same time, although there have been considerable research studies of application workloads based on GPUs and CPUs, our theoretical understanding of the mathematical optimization and performance analysis of resource allocation in high-performance supercomputing environments is far more limited and largely unexplored under QPU-only and QPU-CPU-combined workloads, especially with respect to the latter.

In this paper, we seek to gain such a theoretical understanding of the foundations for the mathematical optimization and performance analysis of resource allocation in quantum-centric supercomputing environments comprising a combination of QPUs, GPUs and CPUs. A key design issue concerns the patterns of application workloads in these high-performance environments that comprise various combinations of core parallel and sequential computational phases across the different processing units. In particular, these two basic core components of quantum-centric supercomputing application workloads consist of (1) quantum and classical computational tasks that are executed simultaneously in parallel on both QPUs and GPUs, and (2) quantum and classical computational tasks that are executed individually in sequence on QPUs and GPUs in any order. Hence another key design issue concerns the performance of these core parallel and sequential computational phases as the building blocks of quantum-centric supercomputing application workloads. Yet another key design issue concerns the resource allocation policies that should be employed in quantum-centric supercomputing environments, where the optimal resource allocation policy is not well understood nor is the performance of a not uncommon reservation-based resource allocation policy relative to the optimal policy.

Our Contributions. Our objective in this paper is to establish a general theoretical foundation for the mathematical optimization and performance analysis of resource allocation in quantum-centric supercomputing environments. This leads to the following important technical contributions. First, we formulate mathematical models of the application workload patterns in these high-performance environments, with an emphasis on jobs requiring combinations of QPUs, GPUs and CPUs, and then focus on a particular instance that spans the combination of parallel and sequential computational phases at the core of these patterns. Second, we derive a mathematical analysis of the core parallel and sequential computational phases of quantum-centric supercomputing application workloads, which includes exact closed-form solutions for the expected sojourn time of the sequential computational phase and highly accurate closed-form approximations for expected sojourn time of the parallel computational phase. The latter theoretical results provide the first closed-form expressions for the expected sojourn time of the corresponding fork-join queueing network with nonhomogeneous service rates, and thus are of interest in their own right. Third, we derive a matrix-analytic analysis of quantum-centric supercomputing environments under a reservation-based resource allocation policy in which the system resources are reserved by jobs in the order of their arrival. Since it is not uncommon to find this approach employed in practice, we include this analysis for comparison against the optimal resource allocation policy. Fourth, we formulate and derive an efficient numerical solution for a general resource

capacity allocation optimization problem within the context of quantum-centric supercomputing environments that differs significantly from the reservation-based resource allocation policy. Finally, we conduct many numerical experiments to empirically evaluate various aspects of the theoretical foundations established in our study. This includes illustrating the performance of core parallel and sequential computational phases of quantum-centric supercomputing application workloads as a function of various system and workload parameters. These empirical results demonstrate the very strong accuracy of our closed-form approximations for the fork-join queueing networks corresponding to the core parallel computational phases, which is of interest in its own right. Our numerical experiments also demonstrate the characteristics of the optimal resource capacity allocation policy as a function of various system and workload parameters, as well as providing performance comparisons between this optimal policy and the reservation-based resource allocation policy under the same resource capacity allocations.

Organization of paper. Section 2 presents our mathematical models of quantum-centric supercomputing system environments and application workloads, followed by a mathematical analysis of the two basic core components of quantum-centric supercomputing application workloads. We then derive in Section 4 a matrix-analytic analysis of the quantum-centric supercomputing environment under a reservation-based resource allocation policy not uncommonly employed in practice, followed by the derivation of a mathematical framework for determining the optimal resource allocation policy. The proofs of our main theoretical results are presented in Section 6, followed by a broad collection of numerical experiments. Concluding remarks are provided in Section 8, with the appendices providing additional technical details.

2 Mathematical Models

The applications of quantum-centric supercomputing environments are generally characterized by a directed acyclic graph. Here, the directed edges from the set of parent nodes to a child node in the graph define the set of tasks associated with the parent nodes that must be completed before the task associated with the child node can start execution. Based on existing and anticipated application patterns [8, 18, 56], the basic core components of these directed acyclic graphs that make up quantum-centric supercomputing application workloads are represented by: (1) a parallel computational phase comprising a single fork-join pair of quantum and classical computational tasks executed in parallel; and (2) a sequential computational phase comprising a single pair of quantum and classical computational tasks executed in series in either order. Then the quantum-centric supercomputing application workloads consist of a mixture of these two basic core components, where quantum computational tasks are executed on system QPUs and classical computational tasks are executed on system GPUs. As a common representative model of these workloads, we shall primarily consider two classes of quantum-centric supercomputing applications with different structures for the computational phases depicted in the directed acyclic graph. The first class, called *parallel-sequential applications*, consists of a parallel computational phase followed by

a sequential computational phase. The second class, called *sequential-parallel applications*, consists of a sequential computational phase followed by a parallel computational phase. Although not a strict requirement, we envision that both application classes start with a classical initialization task executed on system CPUs and end with a classical termination task executed on system CPUs.

A job arrives to the quantum-centric supercomputing environment according to a Poisson process with rate λ . With probability $p_{\parallel}$, the arriving job belongs to the parallel-sequential application class, and with probability $1 - p_{\parallel}$, the arriving job belongs to the sequential-parallel application class. In both application classes, the sequential computational phase consists of a quantum computational task followed by a classical computational task with probability $p_{\circ}$, and consists of a classical computational task followed by a quantum computational task with probability $1 - p_{\circ}$. The quantum computational tasks of any parallel computational phase or sequential computational phase can involve parallel quantum computations requiring execution on $M_Q \geq 1$ QPUs. Similarly, the classical computational tasks of any parallel computational phase or sequential computational phase can involve parallel classical computations requiring execution on $M_G \geq 1$ GPUs. As our starting point, we shall consider $M_Q = 1$ for all parallel computational phases and $M_G = 1$ for all sequential computational phases. While this helps to reduce the complexity of the mathematical analysis and optimization problems at hand, it is important to note that queueing performance such as the expected sojourn time in a system with M_Q QPUs operating at unit speed converges asymptotically to the corresponding queueing performance in our assumed single QPU with speed M_Q in the limit as the system utilization $\rho_Q \rightarrow 1$; and similarly for a system with M_G GPUs operating at unit speed converging asymptotically to our assumed single GPU with speed M_G in the limit as the system utilization $\rho_G \rightarrow 1$ [13, 64].

The service times for a quantum computational task on system QPUs as part of the parallel-sequential application class follow an exponential distribution with rate μ_Q^{PP} ; similarly, the service times for a quantum computational task on system QPUs as part of the sequential-parallel application class follow an exponential distribution with rate μ_Q^{PS} or μ_Q^{SP} , depending upon the order of the series of tasks in the sequential computational phase. Analogously, the service times for a classical computational task on system GPUs as part of the parallel-sequential application class follow an exponential distribution with rate μ_G^{PP} ; similarly, the service times for a classical computational task on system GPUs as part of the sequential-parallel application class follow an exponential distribution with rate μ_G^{PS} or μ_G^{SP} , depending upon the order of the series of tasks in the sequential computational phase. The service times for a classical initialization and termination task on system CPUs follow an exponential distribution with rate μ_I^P, μ_I^S and μ_T^P, μ_T^S , respectively, such that $\mu_I^P, \mu_I^S, \mu_T^P, \mu_T^S \gg \mu_Q^{PP}, \mu_Q^{PS}, \mu_Q^{SP}, \mu_G^{PP}, \mu_G^{PS}, \mu_G^{SP}$. There are three different instances of this quantum-centric supercomputing application workload model. The first, called *balanced application workloads*, consists of homogeneous service rates where $\mu_Q^{PP} = \mu_G^{PP}$, $\mu_Q^{PS} = \mu_G^{PS}$, and $\mu_Q^{SP} = \mu_G^{SP}$; the second instance, called *quantum-dominant application workloads*, consists of nonhomogeneous service rates where $\mu_Q^{PP} < \mu_G^{PP}$, $\mu_Q^{PS} < \mu_G^{PS}$, and $\mu_Q^{SP} < \mu_G^{SP}$; and the third instance, called *classical-dominant application workloads*, consists of nonhomoge-

neous service rates where $\mu_Q^{PP} > \mu_G^{PP}$, $\mu_Q^{PS} > \mu_G^{PS}$, and $\mu_Q^{SP} > \mu_G^{SP}$. Although all three application workload models are considered in our study, it is important to point out that existing and anticipated application patterns suggest that balanced workloads are less likely to be found in quantum-centric supercomputing environments in practice and it is strongly believed that this trend will persist [56]. Quantum-dominant and classical-dominant workloads are expected to be far more prevalent, and thus these two workloads represent an important emphasis in our study.

While our assumptions of Poisson job arrivals and exponential service times help to reduce the complexity of the mathematical analysis and optimization problems at hand, it is critically important to note that there is very little or no data available on the arrival and service patterns for quantum-centric supercomputing environments, and whatever data might be available today is highly unlikely to be representative of future quantum-centric supercomputing application workloads [35]. The field is evolving so quickly with very little in the way of a fundamental analysis of performance in quantum-centric supercomputing environments, making it premature to know the actual distributions of arrival and service patterns in future quantum-centric supercomputing application workloads. Meanwhile, there is an important gap in our understanding of fundamental issues related to design, performance and optimization of quantum-centric supercomputing environments. Given such few studies of this type in the field, we seek to address this gap in the research literature with our study providing the first mathematical analysis and optimization of fundamental performance and related theoretical results in quantum-centric supercomputing environments, to the best of our knowledge.

3 Mathematical Analysis of Core Computational Phases

The two basic core components of quantum-centric supercomputing application workloads consist of either a sequential computational phase or a parallel computational phase. A sequential computational phase comprises with probability p_o a quantum computational task executed on a system QPU followed by a classical computational task executed on a system GPU, and comprises with probability $1 - p_o$ a classical computational task executed on a system GPU followed by a quantum computational task executed on a system QPU. In contrast, a parallel computational phase consists of a single fork-join pair of quantum and classical computational tasks with the former executed on a system QPU and the latter executed on a system GPU. We consider in turn the analysis of each of these two basic core components of quantum-centric supercomputing application workloads in isolation, first deriving a closed-form expressions for the expected sojourn time of the sequential computational phase and then deriving a closed-form expressions for the expected sojourn time of the parallel computational phase.

3.1 Sequential Computational Phase

The stochastic network corresponding to the sequential computational phase of quantum-centric supercomputing application workloads in isolation represents an open Jackson queueing network [6, 26]. Such jobs arrive according to a Poisson process with rate γ , each of which is routed with probability p_o to queue 1 for a system QPU and is routed with probability $1 - p_o$ to queue 2 for a system GPU. Upon completion of the quantum computational task at queue 1, the corresponding jobs are routed directly to queue 3 for a system GPU; and upon completion of the classical computational task at queue 2, the corresponding jobs are routed directly to queue 4 for a system QPU. The arrivals to each queue i follow Poisson processes with rates $\gamma_1 = \gamma_3 = p_o\lambda$ and $\gamma_2 = \gamma_4 = (1 - p_o)\lambda$; and the service times for each queue i follow an exponential distribution with rate $\hat{\mu}_i$, $i \in \{1, \dots, 4\}$, where $\hat{\mu}_1 = S_{Q_1}\mu_{Q_1}$, $\hat{\mu}_2 = S_{G_2}\mu_{G_2}$, $\hat{\mu}_3 = S_{Q_3}\mu_{Q_3}$, $\hat{\mu}_4 = S_{G_4}\mu_{G_4}$. Then, jobs depart the system upon completion of their computational tasks at queue 3 or queue 4. From Burke's Theorem [12] and the quasireversibility of such stochastic networks [27, 39], we also have that the departure process from queue 3 and queue 4 are Poisson with rate $p_o\lambda$ and $(1 - p_o)\lambda$, respectively.

Stochastic networks of this form have received considerable attention in the research literature; see, e.g., [2, 6, 11, 26, 27, 33]. It is generally well known that these open Jackson queueing networks have a product-form solution, which implies that each queue of the stochastic network can be analyzed in a separable manner from the rest of the network. We therefore have that the expected sojourn time at queue i is given by

$$\mathbb{E}[T_{PF}^{(i)}] = \frac{1}{\hat{\mu}_i - \gamma_i}, \quad i \in \{1, \dots, 4\}, \quad (1)$$

and that the expected overall sojourn time of the sequential computational phase of quantum-centric supercomputing application workloads in isolation is obtained as the sum of the expected sojourn time at each of the queues i in (1); namely, we have

$$\mathbb{E}[T_{PF}] = \frac{1}{\hat{\mu}_1 - \gamma_1} + \frac{1}{\hat{\mu}_2 - \gamma_2} + \frac{1}{\hat{\mu}_3 - \gamma_3} + \frac{1}{\hat{\mu}_4 - \gamma_4}. \quad (2)$$

Although we primarily focus on a single pair of alternating quantum and classical computational tasks executed in series in either order, the corresponding results for a general number of sequential computational phases would be obtained in a similar manner.

3.2 Parallel Computational Phase

The stochastic network corresponding to the parallel computational phase of quantum-centric supercomputing application workloads in isolation represents a fork-join queueing network with two parallel servers [16, 17]. Such jobs arrive according to a Poisson process with rate γ , each of which forks into a pair of quantum and classical computational tasks to be executed in parallel with one routed to a queue for the system QPU and the other routed to a queue for the system GPU. The service times for quantum computational tasks at the system QPU queue and for classical computational tasks at the system GPU queue each

follow an exponential distribution with rate $\hat{\mu}_Q = S_Q\mu_Q$ and $\hat{\mu}_G = S_G\mu_G$, respectively. Without loss of generality, let $\hat{\mu}_1 = \min\{\hat{\mu}_Q, \hat{\mu}_G\}$ and $\hat{\mu}_2 = \max\{\hat{\mu}_Q, \hat{\mu}_G\}$, and thus $\hat{\mu}_1 \leq \hat{\mu}_2$. Then, jobs depart the system only upon completion of their computational tasks at both queue 1 and queue 2. Define the critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$. For the purpose of analyzing quantum-dominant and classical-dominant application workloads relative to balanced application workloads, define the heterogeneity ratio $r := \hat{\mu}_2/\hat{\mu}_1 \geq 1$.

Stochastic networks of this form have received considerable attention in the research literature; see, e.g., [5, 16, 17, 37, 41, 47, 53]. In the case of balanced application workloads with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$, it is generally well known that the expected sojourn time of jobs in these fork-join queueing networks is given by

$$\mathbb{E}[T_{FJ}] = \frac{12 - \gamma/\hat{\mu}}{8(\hat{\mu} - \gamma)}, \quad (3)$$

due to Nelson and Tantawi [41]. Hence, the expected overall sojourn time of the parallel computational phase of quantum-centric supercomputing application workloads in isolation can be expressed as in (3).

On the other hand, as discussed in Section 2, the service times of quantum computational tasks and classical computational tasks in quantum-centric supercomputing application workloads are very rarely balanced, if at all, and thus depict either quantum-dominant workloads where $\hat{\mu}_Q < \hat{\mu}_G$ or classical-dominant workloads where $\hat{\mu}_G < \hat{\mu}_Q$. In strong contrast to (3), the corresponding fork-join queueing network under unbalanced workloads with nonhomogeneous service rates generally represents a difficult queueing-theoretic problem for which there is no known exact expression of the expected sojourn time. To address this difficult queueing-theoretic problem, we derive a couple of mathematical approximations based on different forms of interpolation. Our first approximation approach consists of deriving upper and lower bounds and then interpolating between these two bounds. Our second approximation approach consists of deriving a light-heavy traffic interpolation approximation. A more detailed discussion of this latter approach, developed in large part by Reiman and Simon [49–51], is presented in Appendix A to which we refer for additional technical details. We derive in turn each of our closed-form mathematical approximations and related theoretical results for two-server fork-join queueing networks with nonhomogeneous service rates, which are of interest in their own right, noting that both our approximations yield the exact result in (3) for fork-join queueing networks with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$. The numerical results in Section 7 demonstrate that both approaches provide very accurate results for the analysis of otherwise intractable queueing systems.

3.2.1 Upper-Lower Bound Interpolation

Our upper-lower bound interpolation approach is based on a convex combination of an upper and a lower bound on the expected sojourn time in the fork-join queueing network under nonhomogeneous service rates $\hat{\mu}_1 < \hat{\mu}_2$. We start by first deriving the following theoretical result for these upper and lower bounds.

Lemma 1. *The expected sojourn time of the fork-join queueing network $\mathbb{E}[T_{FJ}]$ is upper bounded by*

$$\mathbb{E}[T_{FJ}] \leq \mathbb{E}[T_{FJ}^{UB}] = \frac{1}{\hat{\mu}_1 - \gamma} + \frac{1}{\hat{\mu}_2 - \gamma} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2 - 2\gamma}. \quad (4)$$

Furthermore, the expected sojourn time of the fork-join queueing network is lower bounded by

$$\mathbb{E}[T_{FJ}] \geq \mathbb{E}[T_{FJ}^{LB}] = \frac{1}{\hat{\mu}_1 - \gamma}. \quad (5)$$

Recall $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$ and $r := \hat{\mu}_2/\hat{\mu}_1 \geq 1$. Observe that, in the case of $r = 1$ where $\hat{\mu}_1 = \hat{\mu}_2$, we have $\mathbb{E}[T_{FJ}^{UB}] = (3/2) \cdot \mathbb{E}[T_{MM1}^{(1)}]$ and $\mathbb{E}[T_{FJ}^{LB}] = \mathbb{E}[T_{MM1}^{(1)}]$, resulting in a 50% gap between the upper and lower bounds with the exact expected sojourn time in (3) falling strictly between the two bounds. Moreover, in the light-traffic limit as $\gamma \rightarrow 0$, $\mathbb{E}[T_{FJ}]$ tends toward $\mathbb{E}[T_{FJ}^{UB}]$ with $\mathbb{E}[T_{FJ}]/\mathbb{E}[T_{FJ}^{UB}] \rightarrow 1$; and, in the heavy-traffic limit as $\gamma \rightarrow \hat{\mu}^*$, $\mathbb{E}[T_{FJ}]$ tends toward $\mathbb{E}[T_{FJ}^{LB}]$ with $\mathbb{E}[T_{FJ}]/\mathbb{E}[T_{FJ}^{LB}] \rightarrow 11/8$. On the other hand, we observe that the gap between the upper and lower bounds decreases with increasing $r > 1$. Moreover, in the limit as $r \rightarrow \infty$, we have that $\mathbb{E}[T_{FJ}^{UB}]$ converges from above to $\mathbb{E}[T_{FJ}^{LB}] = 1/(\hat{\mu}_1 - \gamma)$ and thus the bounds become tight asymptotically with the expected sojourn time well approximated by that of the bottleneck M/M/1 queue.

We now exploit the above results to obtain the desired upper-lower bound interpolation approximation for the expected sojourn time of jobs in the two-server fork-join queueing network of interest under unbalanced application workloads.

Theorem 1. *Consider the sojourn time T_{FJ} of jobs in a fork-join queueing network comprising two parallel servers where arrivals follow a Poisson process with rate γ and service times follow an exponential distribution with nonhomogeneous rates $\hat{\mu}_1$ and $\hat{\mu}_2$ at server 1 and 2, respectively. Without loss of generality, let $\hat{\mu}_1 = \mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. The upper-lower bound interpolation approximation is then given by*

$$\mathbb{E}[T_{FJ}] = (1 - \hat{\alpha}) \cdot \left(\frac{1}{\hat{\mu}_1 - \gamma} + \frac{1}{\hat{\mu}_2 - \gamma} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2 - 2\gamma} \right) + \hat{\alpha} \cdot \left(\frac{1}{\hat{\mu}_1 - \gamma} \right), \quad \lambda \in [0, \hat{\mu}_1), \quad (6)$$

where $\hat{\alpha} = (\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)/8$.

Remark 1. *It is important to note that, upon substituting $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}$ in (6), our upper-lower bound interpolation approximation recovers the exact expected sojourn time of jobs in the two-server fork-join queueing network of interest under balanced application workloads, as specifically expressed in (3).*

3.2.2 Light-Heavy Traffic Interpolation

Our light-heavy traffic interpolation approach is based on a combination of light-traffic derivatives associated with the Taylor series expansion of $\mathbb{E}[T_{FJ}]$ at $\gamma = 0$ and the heavy-traffic limit of $\mathbb{E}[T_{FJ}]$ as γ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$. The light-traffic theory rigorously developed in [51] enable us in principle to calculate the derivative

of any order n , thus providing the means to fully determine an exact expression for $\mathbb{E}[T_{FJ}]$ in terms of its Taylor series expansion and the corresponding light-traffic derivatives for all n . However, as is often the case in practice, the calculations of these light-traffic derivatives quickly become extremely involved. We therefore take the approach of combining a few of the light-traffic derivatives with the heavy-traffic limit. In particular, we derive the 0^{th} and 1^{st} order derivatives together with an approximation for the heavy-traffic limit, and then combine these results to obtain our 0^{th} and 1^{st} order light-heavy traffic interpolation approximations.

Consider the sojourn time T_{FJ} of a tagged job that arrives at time $t = 0$ whose service times v_1 and v_2 are random variables that respectively follow an exponential distribution with mean $\mathbb{E}[v_1] = \hat{\mu}_1$ and $\mathbb{E}[v_2] = \hat{\mu}_2$, where v_1 and v_2 are independent of all else. The 0^{th} order derivative can be obtained by definition as

$$\mathbb{E}[\psi(\emptyset)] := \mathbb{E}[T_{FJ} \mid \text{no arrivals over } (-\infty, 0)], \quad (7)$$

noting that any arrival at time $t \in (0, \infty)$ will be serviced after the tagged job and therefore not impact T_{FJ} . We then have the following theoretical result for $\mathbb{E}[\psi(\emptyset)]$.

Lemma 2. *The 0^{th} order derivative of the fork-join queueing network is given by*

$$\mathbb{E}[\psi(\emptyset)] = \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \quad (8)$$

Turning to the 1^{st} order derivative, we consider the sojourn time T_{FJ} of a tagged job that arrives at time 0 with exponential service times v_1 and v_2 as above, given that another job arrives at time $t \in (-\infty, \infty)$ whose service times w_1 and w_2 are random variables that respectively follow an exponential distribution with mean $\mathbb{E}[w_1] = \hat{\mu}_1$ and $\mathbb{E}[w_2] = \hat{\mu}_2$, where v_1, v_2, w_1, w_2 are independent of all else. The 1^{st} order derivative can be obtained by definition as

$$\mathbb{E}[\psi(\{t\})] := \mathbb{E}[T_{FJ} \mid \text{exactly one arrival at time } t \text{ over } (-\infty, \infty)], \quad (9)$$

$$\left. \frac{d\mathbb{E}[T_{FJ}]}{d\lambda} \right|_{\lambda=0} = \int_{-\infty}^{\infty} (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt. \quad (10)$$

We then have the following theoretical result for 1^{st} order derivative.

Lemma 3. *The 1^{st} order derivative of the fork-join queueing network is given by*

$$\left. \frac{d\mathbb{E}[T_{FJ}]}{d\lambda} \right|_{\lambda=0} = \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4}. \quad (11)$$

Recall $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$ and $r := \hat{\mu}_2/\hat{\mu}_1 \geq 1$. Let us next consider the heavy-traffic limit of the expected sojourn time $\mathbb{E}[T_{FJ}]$ of jobs in the two-server fork-join queueing network under any given $\hat{\mu}_1$ and $\hat{\mu}_2$ as γ approaches its critical value μ^* , namely

$$\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma) \mathbb{E}[T_{FJ}] = h^*. \quad (12)$$

Our approach consists of first establishing upper and lower bounds on the heavy-traffic limit h^* as $r \rightarrow 1$ and as $r \rightarrow \infty$, and then obtaining an upper-lower bound interpolation approximation for h^* . The corresponding theoretical results are presented as follows.

Lemma 4. *The heavy-traffic limit of the fork-join queueing network is bounded above and below by*

$$\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma) \mathbb{E}[T_{FJ}] = h^* \in [1, 11/8]. \quad (13)$$

Furthermore, the upper-lower bound interpolation approximation for this heavy-traffic limit is given by

$$h^* \approx h(r) = 1 + \frac{3}{8} r^{-\beta} \in [1, 11/8], \quad r \in [1, \infty), \quad \beta \in (0, \infty). \quad (14)$$

The upper-lower bound interpolation approximation $h(r)$ decays more rapidly under larger control parameters β for a given heterogeneity ratio r . Simulation results strongly suggest that the transition from $h(r)$ around the upper bound of $11/8$ toward the lower bound of 1 for a given $r \in (1, \infty)$ is very sharp. For example, the heavy-traffic limit $h(r)$ has already decayed to approximately 1.008 for $r = 1.5$. Matching this estimate of $h(1.5) \approx 1.008$ to (14) yields $\beta = \log(0.375/0.008)/\log(1.5) \approx 9.5$.

We now exploit the above results to obtain the desired two light-heavy traffic interpolation approximations $\tilde{f}_0(\gamma)$ and $\tilde{f}_1(\gamma)$ for the expected sojourn time of jobs in the two-server fork-join queueing network of interest under unbalanced application workloads.

Theorem 2. *Consider the sojourn time T_{FJ} of jobs in a fork-join queueing network comprising two parallel servers where arrivals follow a Poisson process with rate γ and service times follow an exponential distribution with nonhomogeneous rates $\hat{\mu}_1$ and $\hat{\mu}_2$ at server 1 and 2, respectively. Further let $f(\gamma) = \mathbb{E}[T_{FJ}]$ and, without loss of generality, $\hat{\mu}_1 = \mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. The 0th and 1st order light-heavy traffic interpolation approximations $\tilde{f}_0(\gamma)$ and $\tilde{f}_1(\gamma)$ for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1)$, are then given by*

$$\tilde{f}_0(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \frac{h(r)}{\hat{\mu}_1} + \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)}, \quad (15)$$

$$\tilde{f}_1(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2 \hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)}. \quad (16)$$

Remark 2. *It is important to note that, upon substituting $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}$ in (15) and (16), our light-heavy traffic interpolation approximations recover the exact expected sojourn time of jobs in the two-server fork-join queueing network of interest under balanced application workloads, as specifically expressed in (3).*

4 Mathematical Analysis of Reserved Allocation

We now consider our quantum-centric supercomputing application and system models of Section 2 under a resource allocation reservation policy in which the resources of the quantum-centric supercomputing environment are reserved by jobs in the order of their arrival. This is not an uncommon scheme in practice today for many quantum-centric supercomputing environments. Recall that each arriving job belongs to the parallel-sequential application class with probability $p_{\parallel}$ and belongs to the sequential-parallel application class with probability $1 - p_{\parallel}$, where the former comprises a parallel computational phase followed by a sequential computational phase and the latter comprises a sequential computational phase followed by a parallel computational phase. The sequential computational phase of both application classes consists of a quantum computational task followed by a sequential computational task with probability p_o , and consists of a sequential computational task followed by a quantum computational task with probability $1 - p_o$. In addition, an arriving job of each application class starts with a classical initialization task and ends with a classical termination task. Due to the not uncommonly employed reservation policy of this section, each job reserves the system CPUs, QPUs and GPUs throughout its execution without any overlap with other jobs that wait in the system queue.

More formally, consider a continuous-time Markov process $\{X(t); t \in \mathbb{R}_+\}$ on the state space $\hat{\Omega} = \bigcup_{i=0}^{\infty} \hat{\Omega}_i$ where $\hat{\Omega}_0 := \{0\}$ and $\hat{\Omega}_i := \{(i, j_1, j_2, k, q, g) \mid i \in \mathbb{Z}_+, j_1 \in \{P, S\}, j_2 \in \{P, S, S_{gq}, S_{qg}\}, k \in \{I, C, T\}, q \in \{0, 1\}, g \in \{0, 1\}\}$, for the nonnegative reals $\mathbb{R}_+ := \mathbb{R}_{\geq 0}$ with $\mathbb{R}_{\geq k} := [k, \infty)$, and nonnegative integers $\mathbb{Z}_+ := \mathbb{Z}_{\geq 0}$ with $\mathbb{Z}_{\geq k} := \{k, k+1, k+2, \dots\}$. Here, the state-vector variable i denotes the total number of jobs in the quantum-centric supercomputing environment, j_1 denotes whether the job in service belongs to the parallel-sequential application class (P) or the sequential-parallel application class (S), j_2 denotes whether the application in service is executing a parallel computational phase (P) or a sequential computational phase (S) that comprises a classical computational task followed by a quantum computational task (S_{gq}) or a quantum computational task followed by a classical computational task (S_{qg}), k denotes whether the job in service is currently executing its initialization phase (I) or termination phase (T) on the system CPUs or executing its quantum/classical computational phases (C), q denotes whether or not the job in service is currently executing a quantum computational task on the system QPUs, and g denotes whether or not the job in service is currently executing a classical computational task on the system GPUs. We shall refer to the set $\hat{\Omega}_i$ as level i .

Define $[M] := \{1, \dots, M\}$, $M < \infty$. Let $x_{i,m} \in \hat{\Omega}_i$ be a pseudo-lexicographic ordering of the elements of $\hat{\Omega}_i$, $i \in \mathbb{Z}_{\geq 1}$, $m \in [M]$; similarly, let $\hat{\Omega}_0 = \{x_0\}$. The ordering $x_{i,m}$ for level $i \in \mathbb{Z}_{\geq 1}$ is then given by: $(i, P, P, I, 0, 0)$; $(i, P, P, C, 0, 1)$; $(i, P, P, C, 1, 0)$; $(i, P, P, C, 1, 1)$; $(i, P, S_{gq}, C, 0, 1)$; $(i, P, S_{gq}, C, 1, 0)$; $(i, P, S_{gq}, C, 0, 1)$; $(i, P, S_{gq}, C, 1, 0)$; $(i, P, P, T, 0, 0)$; $(i, S, S, I, 0, 0)$; $(i, S, P, C, 0, 1)$; $(i, S, P, C, 1, 0)$; $(i, S, P, C, 1, 1)$; $(i, S, S_{gq}, C, 0, 1)$; $(i, S, S_{gq}, C, 1, 0)$; $(i, S, S_{gq}, C, 0, 1)$; $(i, S, S_{gq}, C, 1, 0)$; and thus $M = 18$. Following this pseudo-lexicographic ordering, the infinitesimal generator matrix $\mathbf{Q}$ of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ has the

block tridiagonal structure

$$\mathbf{Q} = \begin{bmatrix} \mathbf{B}_{00} & \mathbf{B}_{01} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{B}_{10} & \mathbf{B}_{11} & \mathbf{A}_1 & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{A}_{-1} & \mathbf{A}_0 & \mathbf{A}_1 & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{-1} & \mathbf{A}_0 & \mathbf{A}_1 & \ddots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}, \quad (17)$$

where $\mathbf{B}_{00}$, $\mathbf{B}_{01}$, $\mathbf{B}_{10}$, $\mathbf{B}_{11}$ and $\mathbf{A}_i$, $i \in \{-1, 0, 1\}$, are finite matrices of dimensions 1×1 , $1 \times M$, $M \times 1$, $M \times M$ and $M \times M$, respectively. By definition, since $\mathbf{Q}$ is a generator matrix, we have that $\mathbf{B}_{01}$, $\mathbf{B}_{10}$, $\mathbf{A}_{-1}$ and $\mathbf{A}_1$ are nonnegative matrices while $\mathbf{B}_{00}$, $\mathbf{B}_{11}$ and $\mathbf{A}_0$ have nonnegative offdiagonal elements and strictly negative diagonal elements such that the row sums of $\mathbf{Q}$ equal 0. Moreover, we have that $\mathbf{B}_0 = \mathbf{B}_{00} + \mathbf{B}_{01}$, $\mathbf{B}_1 = \mathbf{B}_{10} + \mathbf{B}_{11} + \mathbf{A}_1$, and $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ are generator matrices.

The matrices corresponding to the non-homogeneous boundary of the state space $\hat{\Omega}$ are given by

$$\mathbf{B}_{00} = [-\lambda], \quad \mathbf{B}_{01} = [\mathbf{B}_{01}^P \quad \mathbf{B}_{01}^S], \quad \mathbf{B}_{10} = [\mathbf{B}_{10}^P \quad \mathbf{B}_{10}^S]^\top, \quad (18)$$

$$\mathbf{B}_{01}^P = [p_{\parallel}\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad \mathbf{B}_{01}^S = [(1-p_{\parallel})\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad (19)$$

$$\mathbf{B}_{10}^P = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^P], \quad \mathbf{B}_{10}^S = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^S], \quad (20)$$

$$\mathbf{B}_{11} = \begin{bmatrix} \mathbf{B}_{11}^P & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{11}^S \end{bmatrix}, \quad \mathbf{B}_{11}^P = \begin{bmatrix} \mathbf{B}_{11}^{P_{pp}} & \mathbf{B}_{11}^{P_{ps}} \\ \mathbf{0} & \mathbf{B}_{11}^{P_{ss}} \end{bmatrix}, \quad \mathbf{B}_{11}^S = \begin{bmatrix} \mathbf{B}_{11}^{S_{pp}} & \mathbf{B}_{11}^{S_{ps}} \\ \mathbf{B}_{11}^{S_{sp}} & \mathbf{B}_{11}^{S_{ss}} \end{bmatrix}, \quad (21)$$

where $\mathbf{B}_{11}^{P_{pp}}$, $\mathbf{B}_{11}^{P_{ps}}$, $\mathbf{B}_{11}^{P_{ss}}$, $\mathbf{B}_{11}^{S_{pp}}$, $\mathbf{B}_{11}^{S_{ps}}$, $\mathbf{B}_{11}^{S_{sp}}$ and $\mathbf{B}_{11}^{S_{ss}}$ are given in Appendix B together with additional details. Similarly, the matrices for the homogeneous portion of the state space are given by $\mathbf{A}_1 = \lambda \mathbf{I}$, $\mathbf{A}_0 = \mathbf{B}_{11}$, and

$$\mathbf{A}_{-1} = \begin{bmatrix} \mathbf{A}_{-1}^{PP} & \mathbf{A}_{-1}^{PS} \\ \mathbf{A}_{-1}^{SP} & \mathbf{A}_{-1}^{SS} \end{bmatrix}, \quad (22)$$

where $\mathbf{I}$ denotes the identity matrix of appropriate dimension, and where $\mathbf{A}_{-1}^{PP}$, $\mathbf{A}_{-1}^{PS}$, $\mathbf{A}_{-1}^{SP}$ and $\mathbf{A}_{-1}^{SS}$ are given in Appendix B together with additional details.

Following the pseudo-lexicographic ordering of the elements of each level i of the state space, define

$$\boldsymbol{\pi} := (\boldsymbol{\pi}_0, \boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \dots), \quad \boldsymbol{\pi}_0 := (\hat{\pi}(x_0)), \quad \hat{\pi}(x_0) := \lim_{t \rightarrow \infty} \mathbb{P}[X(t) = x_0], \quad (23)$$

$$\boldsymbol{\pi}_i := (\hat{\pi}(x_{i,1}), \dots, \hat{\pi}(x_{i,M})), \quad \hat{\pi}(x_{i,m}) := \lim_{t \rightarrow \infty} \mathbb{P}[X(t) = x_{i,m}], \quad i \in \mathbb{Z}_{\geq 1}, \quad m \in [M]. \quad (24)$$

Assuming the Markov process $\{X(t); t \in \mathbb{R}_+\}$ to be irreducible and ergodic, the limiting probability vector $\boldsymbol{\pi}$ is the stationary distribution of this stochastic process such that its invariant probability vector $\boldsymbol{\pi}$ exists and is uniquely determined as the solution of $\boldsymbol{\pi} = \boldsymbol{\pi} \mathbf{P}$

and $\boldsymbol{\pi}\mathbf{e} = 1$, where $\mathbf{e} = (1, \dots, 1)^\top$ [11, 34]. Let $\text{sp}(\mathbf{Y})$ denote the spectral radius of a matrix $\mathbf{Y}$, and let a solution $\mathbf{Y}$ be called a minimal nonnegative solution of a matrix equation if $\mathbf{0} \leq \mathbf{Y} \leq \mathbf{Z}$ for any other nonnegative solution $\mathbf{Z}$, with the inequalities holding componentwise. The stationary probability vector $\boldsymbol{\pi}$ of the process $\{X(t); t \in \mathbb{R}_+\}$ with generator $\mathbf{Q}$ then can be obtained from the following matrix-analytic results.

Theorem 3 (Theorems 1.7.1 and 1.5.1 [43]). *Suppose that the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible and aperiodic with $\mathbf{Q}$ in the form of (17). Further suppose the generator matrix $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ is irreducible. Then this stochastic process is positive recurrent if and only if $\text{sp}(\mathbf{R}) < 1$, where $\mathbf{R}$ is the minimal nonnegative solution of the quadratic matrix equation*

$$\mathbf{R}^2\mathbf{A}_{-1} + \mathbf{R}\mathbf{A}_0 + \mathbf{A}_1 = \mathbf{0}. \quad (25)$$

Furthermore, there exists a positive vector $(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ such that

$$(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1) \begin{bmatrix} \mathbf{B}_{00} & \mathbf{B}_{01} \\ \mathbf{B}_{10} & \mathbf{B}_{11} + \mathbf{R}\mathbf{A}_{-1} \end{bmatrix} = \mathbf{0}. \quad (26)$$

The remaining elements of the invariant probability vector $\boldsymbol{\pi}$ are then given by

$$\boldsymbol{\pi}_{n+1} = \boldsymbol{\pi}_1 \mathbf{R}^n, \quad n \in \mathbb{Z}_+, \quad (27)$$

where $(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ is normalized by

$$\boldsymbol{\pi}_0\mathbf{e} + \boldsymbol{\pi}_1(\mathbf{I} - \mathbf{R})^{-1}\mathbf{e} = 1. \quad (28)$$

We have above assumed the irreducibility and ergodicity of the Markov process $\{X(t); t \in \mathbb{R}_+\}$. It readily follows from the definition of $\{X(t); t \in \mathbb{R}_+\}$ and its generator matrix $\mathbf{Q}$ that the Markov process is irreducible and aperiodic. Then the stability conditions for the Markov process to be positive recurrent are provided in Theorem 3, under the additional assumption that the generator matrix $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ is irreducible. The irreducibility of $\mathbf{A}$ can be readily verified, as the directed graph corresponding to $\mathbf{A}$ is strongly connected with transitions from state $(*, P, P, I, 0, 0)$ to state $(*, P, P, T, 0, 0)$ reaching every state $(*, P, *, C, *, *)$ in between, transitions from state $(*, S, S, I, 0, 0)$ to state $(*, S, S, T, 0, 0)$ reaching every state $(*, S, *, C, *, *)$ in between, and transitions from states $(*, *, *, T, 0, 0)$ to states $(*, *, *, I, 0, 0)$ connecting the states across both application classes. Since the generator matrix $\mathbf{A}$ of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible, we then have necessary and sufficient conditions for $\text{sp}(\mathbf{R}) < 1$ from the following matrix-analytic result. The stochastic process is therefore positive recurrent if and only if (29) is satisfied.

Theorem 4 (Theorems 1.7.1 [43]). *Suppose that the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible. Further suppose that the generator $\mathbf{A}$ is irreducible. Then $\text{sp}(\mathbf{R}) < 1$ if and only if*

$$\mathbf{y}\mathbf{A}_1\mathbf{e} < \mathbf{y}\mathbf{A}_{-1}\mathbf{e}, \quad (29)$$

where $\mathbf{y}$ is the stationary probability vector of $\mathbf{A}$, i.e., the (unique) solution of $\mathbf{y}\mathbf{A} = \mathbf{0}$ and $\mathbf{y}\mathbf{e} = 1$.

Once the stationary distribution is determined, various functions of the elements of the invariant probability vector $\boldsymbol{\pi}$ can be used to obtain performance measures and quantities of interest. In particular, the tail distribution of the queue length process can be expressed as

$$\mathbb{P}[Q > z] = \sum_{n=z+1}^{\infty} \boldsymbol{\pi}_n \mathbf{e} = \boldsymbol{\pi}_1 \mathbf{R}^z (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e}, \quad z \in \mathbb{Z}_+, \quad (30)$$

with the corresponding expectation given by

$$\mathbb{E}[Q] = \sum_{n=0}^{\infty} (n+1) \boldsymbol{\pi}_{n+1} \mathbf{e} = \boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e} + \boldsymbol{\pi}_1 \mathbf{R} (\mathbf{I} - \mathbf{R})^{-2} \mathbf{e}. \quad (31)$$

The expected sojourn time of a job can then be calculated using Little's law [66] and (31), which yields

$$\mathbb{E}[T] = \lambda^{-1} (\boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e} + \boldsymbol{\pi}_1 \mathbf{R} (\mathbf{I} - \mathbf{R})^{-2} \mathbf{e}). \quad (32)$$

The asymptotic decay rate of the tail distribution of the queue length process is another measure of interest. Let η denote this decay rate, which is also called the caudal characteristic [45, 46]. Due to the irreducibility of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ and the irreducibility of the generator matrix $\mathbf{A}$, we either have $\eta = \text{sp}(\mathbf{R})$ when the matrix $\mathbf{R}$ is irreducible, or otherwise have $\eta = \text{sp}(\mathbf{R}_{11})$ where, after appropriate row and column permutations,

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{R}_{22} \end{bmatrix}$$

with $\mathbf{R}_{11}$ irreducible and $\mathbf{R}_{22}$ strictly upper triangular. We then conclude that $\mathbf{R}^z = O(\eta^z)$, as $z \rightarrow \infty$. Let $\mathbf{u}$ and $\mathbf{v}$ be the left and right eigenvectors corresponding to η that are strictly positive and normalized by $\mathbf{u}\mathbf{e} = 1$ and $\mathbf{u}\mathbf{v} = 1$. From the properties of the Markov process $\{X(t); t \in \mathbb{R}_+\}$, we know that [55]

$$\mathbf{R}^z = \eta^z \mathbf{v} \cdot \mathbf{u} + o(\eta^z), \quad \text{as } z \rightarrow \infty,$$

which together with equation (27) yields

$$\boldsymbol{\pi}_{z+1} \mathbf{e} = \boldsymbol{\pi}_1 \mathbf{v} \eta^z + o(\eta^z), \quad \text{as } z \rightarrow \infty. \quad (33)$$

It then follows that

$$\mathbb{P}[Q \geq z+1] = \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1 - \eta} \eta^z + o(\eta^z), \quad \text{as } z \rightarrow \infty, \quad (34)$$

and thus

$$\lim_{z \rightarrow \infty} \frac{\mathbb{P}[Q \geq z+1]}{\eta^z} = \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1 - \eta}, \quad (35)$$

or equivalently

$$\mathbb{P}[Q \geq z+1] \sim \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1 - \eta} \eta^z, \quad \text{as } z \rightarrow \infty, \quad (36)$$

where $f(z) \sim g(z)$ denotes that $\lim_{z \rightarrow \infty} f(z)/g(z) = 1$. Hence, in addition to providing the stability criterion for the quantum-centric supercomputing environment, η is indicative of the tail behavior of the stationary queue length distribution.

Note that the caudal characteristic can be obtained without having to first solve for the matrix $\mathbf{R}$ in (25). Define the matrix $\tilde{\mathbf{A}}(s) := \mathbf{A}_1 + s\mathbf{A}_0 + s^2\mathbf{A}_{-1}$, for $0 < s \leq 1$. When the generator matrix $\mathbf{A}$ is irreducible, this matrix $\tilde{\mathbf{A}}(s)$ is irreducible with nonnegative off-diagonal elements. Let $\tilde{\eta}(s)$ denote the spectral radius of the matrix $\tilde{\mathbf{A}}(s)$. Then, under the above assumptions, η is the unique solution in $(0,1)$ of the equation $\tilde{\eta}(s) = 0$. This solution can be directly computed, although more efficient methods have been developed [7].

Another set of performance metrics of interest concern the stationary utilization of the system CPUs, QPUs and GPUs composing the quantum-centric supercomputing environment. For each level i , $i \in \mathbb{Z}_+$, the states with elements $k \in \{I, T\}$ represent execution on the system CPUs, the states with elements $q = 1$ represent execution on the system QPUs, and the states with elements $g = 1$ represent execution on the system GPUs. The corresponding stationary utilizations can then be expressed as

$$U_C = \sum_{n=0}^{\infty} \pi_{n+1} \nu_C = \pi_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_C, \quad U_Q = \sum_{n=0}^{\infty} \pi_{n+1} \nu_Q = \pi_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_Q, \quad U_G = \sum_{n=0}^{\infty} \pi_{n+1} \nu_G = \pi_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_G \quad (37)$$

respectively where the ℓ^{th} element of the vector ν_C contains a 1 if $k \in \{I, T\}$ in the corresponding state of the homogeneous portion and contains a 0 otherwise, the ℓ^{th} element of the vector ν_Q contains a 1 if $q = 1$ in the corresponding state of the homogeneous portion and contains a 0 otherwise, and the ℓ^{th} element of the vector ν_G contains a 1 if $g = 1$ in the corresponding state of the homogeneous portion and contains a 0 otherwise.

5 Optimization of Resource Allocation

In contrast to the previous section, as an alternative to the resource allocation reservation policy in fairly common practice today for many quantum-centric supercomputing environments, we shall consider in this section a stochastic network that explicitly represents the queues associated with the quantum-centric supercomputing application and system models of Section 2. A primary goal of this paper is to determine the optimal allocation of resource capacity within the context of quantum-centric supercomputing environments. The fundamental tradeoff here is between the performance received by jobs submitted to the high-performance computing environment and the financial concerns incurred by the host of the environment. We shall focus on the expected sojourn times at the queues of the stochastic network as the performance metric of interest and shall focus on the costs of the resource capacities as the financial metric of interest, noting that related performance and financial metrics can be addressed in a similar manner. Since some queues of the stochastic network may be more important than others, we shall also allow a weight to be assigned to each queue. Then, our formulation of the resource capacity allocation optimization problem of interest consists of minimizing the sum of the weighted expected sojourn times in the

stochastic network subject to a budgetary cost constraint on the resource capacities and to the constraint that the stochastic network is stable. Alternatively, we could similarly consider an optimization formulation based on minimizing the cost of the resource capacities subject to a constraint on the sum of the weighted expected sojourn times.

More formally, let N denote the number of queues comprising the stochastic network with any ordering of interest for these queues. Define the vectors $\boldsymbol{\gamma} := (\gamma_1, \dots, \gamma_N)$ and $\hat{\boldsymbol{\mu}} := (\hat{\mu}_1, \dots, \hat{\mu}_N)$, where γ_i is the effective arrival rate to queue i and $\hat{\mu}_i$ is the resource capacity of queue i . Let $\mathcal{C}$ denote the total budgetary cost constraint on the resource capacities, and $\mathbb{E}[T_i]$ the expected sojourn time at queue i , $i \in [N]$. Further define the vectors $\mathbf{c} := (c_1, \dots, c_N)$ and $\boldsymbol{\omega} := (\omega_1, \dots, \omega_N)$, where c_i is the cost associated with each unit of resource capacity $\hat{\mu}_i$ at queue i and ω_i is the weight associated with the expected sojourn time at queue i , $i \in [N]$. Our resource capacity allocation optimization problem is then formulated as follows.

$$\text{(RCA - OPT)} \quad \min_{\hat{\boldsymbol{\mu}} \in (0, \infty)^N} \sum_{i=1}^N \omega_i \mathbb{E}[T_i] \quad (38)$$

$$\text{s.t.} \quad \sum_{i=1}^N c_i \hat{\mu}_i \leq \mathcal{C}, \quad \text{and} \quad \hat{\mu}_i > \gamma_i, \quad \forall i \in [N]. \quad (39)$$

The difficulty in solving (RCA-OPT) is rooted in determining the expected sojourn times $\mathbb{E}[T_i]$ as a function of the resource capacity vector $\hat{\boldsymbol{\mu}}$. As described in Section 3.1, we know that the sequential computational phase represents a separable product-form queueing network. In addition, a sequential computational phase followed by the fork-join queueing network of a parallel computational phase would also represent a separable queueing network, given the Poisson departure process from the sequential computational phase due to Burke's Theorem [12] and quasireversibility [27, 39]; see Section 3.1. However, even though the service times at each queue follow an exponential distribution, the application workload model comprising combinations of both the parallel-sequential application class and the sequential-parallel application class does not ensure Poisson arrival processes to each queue of the stochastic network. The main difficulty is not knowing the higher-order properties of the arrival process to each queue i beyond its effective arrival γ_i ; moreover, these properties of the arrival process depend on the resource capacities of the queues j that precede queue i in the stochastic network, $i, j \in [N]$. Hence, a standard approach for solving optimization problems such as (RCA-OPT) is based on a combination of simulation methods (to estimate the $\mathbb{E}[T_i]$ for a given $\hat{\boldsymbol{\mu}}$) and gradient methods (to obtain the optimal service rates), such as stochastic approximation [3, 31, 40].

To address the computational complexity of stochastic approximation, we follow the approach of Dieker, Ghosh and Squillante [14] within the context of quantum-centric supercomputing environments. Observe from (1) and (3), and even (49) and (50) in the proof of Theorem 2, that the expression for the expected sojourn time of queue i under any resource capacity allocation $\hat{\boldsymbol{\mu}}$ has a functional form $\mathbb{E}[T_i] = \zeta_i / (\hat{\mu}_i - \gamma_i)$, $i \in [N]$. Upon substituting this functional form into the objective function (38) of (RCA-OPT), we obtain the revised

optimization problem formulation

$$\text{(RCA - OPT : FF)} \quad \min_{\hat{\boldsymbol{\mu}} \in (0, \infty)^N} \sum_{i=1}^N \omega_i \frac{\zeta_i}{\hat{\mu}_i - \gamma_i} \quad (40)$$

$$\text{s.t.} \quad \sum_{i=1}^N c_i \hat{\mu}_i \leq \mathcal{C}, \quad \text{and} \quad \hat{\mu}_i > \gamma_i, \quad \forall i \in [N]. \quad (41)$$

Then, given the vector $\boldsymbol{\zeta} := (\zeta_1, \dots, \zeta_N)$, (RCA-OPT:FF) can be directly solved as established in the following theoretical result.

Theorem 5. *The solution of the resource capacity allocation optimization problem formulation (RCA-OPT:FF) is given by the optimal resource capacity vector $\hat{\boldsymbol{\mu}}^* := (\hat{\mu}_1^*, \dots, \hat{\mu}_N^*)$ where*

$$\hat{\mu}_i^* = \gamma_i + \left(\mathcal{C} - \sum_{j=1}^N c_j \gamma_j \right) \frac{\sqrt{\omega_i \zeta_i / c_i}}{\sum_{j=1}^N \sqrt{\omega_j \zeta_j c_j}}. \quad (42)$$

A similar result was obtained by Kleinrock in the context of multiple channel systems [29] and by Wein in the context of the Brownian approximation of generalized Jackson networks [63].

Based on the result of Theorem 5, we consider the following fixed-point iterative process for the solution of (RCA-OPT:FF). After initializing the resource capacity vector $\hat{\boldsymbol{\mu}}^{(0)}$, we simulate the stochastic queueing network to obtain the expected sojourn times $\mathbb{E}[T_i^{(0)}]$ for each queue $i \in [N]$ under the vector $\hat{\boldsymbol{\mu}}^{(0)}$. The functional form for $\mathbb{E}[T_i^{(0)}]$ is then inverted to obtain $\zeta_i^{(0)} = \mathbb{E}[T_i^{(0)}](\hat{\mu}_i^{(0)} - \gamma_i)$, $i \in [N]$. From (42), we compute the optimal resource capacity vector $\hat{\boldsymbol{\mu}}^{(j)}$ for the next fixed-point iteration $j \in \mathbb{Z}_+$ with iterates given by

$$\hat{\mu}_i^{(j)} = \gamma_i + \left(\mathcal{C} - \sum_{k=1}^N c_k \gamma_k \right) \frac{\sqrt{\omega_i \zeta_i^{(j-1)} / c_i}}{\sum_{k=1}^N \sqrt{\omega_k \zeta_k^{(j-1)} c_k}}, \quad i \in [N], \quad j \in \mathbb{Z}_+. \quad (43)$$

The simulation of the stochastic queueing network under the resource capacity vector $\hat{\boldsymbol{\mu}}^{(j)}$ then yields the corresponding expected sojourn time $\mathbb{E}[T_i^{(j)}]$ for each queue $i \in [N]$, from which the functional form for $\mathbb{E}[T_i^{(j)}]$ yields

$$\zeta_i^{(j)} = \mathbb{E}[T_i^{(j)}](\hat{\mu}_i^{(j)} - \gamma_i), \quad i \in [N], \quad j \in \mathbb{Z}_+. \quad (44)$$

This fixed-point iterative process (43), (44) is repeated until the fixed-point solution converges, the existence of which follows directly from Brouwer's fixed-point theorem [25].

While Dieker, Ghosh and Squillante [14] analyze certain properties of a similar fixed-point iterative process within the context of Brownian tree networks, we turn to consider theoretical properties of our fixed-point iterative process based on (43) within the context of our stochastic queueing network comprising the quantum-centric supercomputing application and system models of Section 2.

6 Proofs of Main Theoretical Results

In this section, we provide the proofs of our main theoretical results throughout the paper, with additional details and more complete proofs provided in the appendices.

6.1 Proof of Lemma 1

The sojourn times $T_{FJ}^{(1)}$ and $T_{FJ}^{(2)}$ of the first and second tasks of arriving jobs share a common Poisson arrival process, and thus they are positively correlated. It is well known that positive correlation reduces the expected maximum $\mathbb{E}[\max\{T_{FJ}^{(1)}, T_{FJ}^{(2)}\}]$ below what it would be if $T_{FJ}^{(1)}$ and $T_{FJ}^{(2)}$ were independent [4]. Hence, upon considering $T_{FJ}^{(1)}$ and $T_{FJ}^{(2)}$ as independent exponential random variables respectively with rates $\hat{\mu}_1 - \gamma$ and $\hat{\mu}_2 - \gamma$ representing the sojourn time of independent M/M/1 queues, we obtain the upper bound in (4) due to the known relationship $\mathbb{E}[\max\{Z_1, Z_2\}] = \mathbb{E}[Z_1] + \mathbb{E}[Z_2] - \mathbb{E}[\min\{Z_1, Z_2\}]$ for independent exponential random variables Z_1 and Z_2 .

Turning to the lower bound, consider the expected sojourn time of the M/M/1 queue with arrival rate γ and service rate $\hat{\mu}_i$ which is given by $\mathbb{E}[T_{MM1}^{(i)}] = 1/(\hat{\mu}_i - \gamma)$, $i = 1, 2$. It follows from Jensen's inequality and the convexity of the maximum function that the expected sojourn time of the fork-join queueing network $\mathbb{E}[T_{FJ}]$ is bounded below by $\max\{\mathbb{E}[T_{MM1}^{(1)}], \mathbb{E}[T_{MM1}^{(2)}]\}$, which represents the expected sojourn time of the bottleneck M/M/1 queue and yields the lower bound in (5).

6.2 Proof of Theorem 1

The upper-lower bound interpolation approximation is based on a convex combination of the upper and lower bounds. Specifically, consider the convex combination

$$\mathbb{E}[T_{FJ}] = (1 - \alpha)\mathbb{E}[T_{FJ}^{UB}] + \alpha\mathbb{E}[T_{FJ}^{LB}] \quad (45)$$

where we seek to determine α from the known exact result for homogeneous service rates. When $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$, we have $\mathbb{E}[T_{FJ}^{UB}] = (3/2) \cdot \mathbb{E}[T_{MM1}^{(1)}]$ and $\mathbb{E}[T_{FJ}^{LB}] = \mathbb{E}[T_{MM1}^{(1)}]$, which yields the following relationship between the exact result for $\mathbb{E}[T_{FJ}]$ in (3) and the convex combination for $\mathbb{E}[T_{FJ}]$ in (45) with respect to α :

$$\frac{12 - \gamma/\hat{\mu}}{8(\hat{\mu} - \gamma)} = (1 - \alpha)(3/2)\mathbb{E}[T_{MM1}^{(1)}] + \alpha\mathbb{E}[T_{MM1}^{(1)}] = \frac{(1 - \alpha)(3/2)}{\hat{\mu} - \gamma} + \frac{\alpha}{\hat{\mu} - \gamma}.$$

Then, upon solving for α , we obtain

$$\alpha = 3 - \frac{12 - \gamma/\hat{\mu}}{4} = \frac{\gamma/\hat{\mu}}{4}.$$

For the fork-join queueing network of interest with nonhomogeneous rates, a natural generalization is to replace $\gamma/\hat{\mu}$ above with the average utilization $(\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)/2$, thus yielding

$$\hat{\alpha} = \frac{(\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)/2}{4} = \frac{(\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)}{8}.$$

Upon substituting (4) and (5) together with the above expression for $\hat{\alpha}$ into (45), we have the desired result in (6), thus completing the proof.

6.3 Proof of Lemma 2

Since the tagged job does not encounter any prior arrivals, we have by definition that $\mathbb{E}[\psi(\emptyset)] = \mathbb{E}[\max\{v_1, v_2\}]$ where the random variable $\psi(\emptyset) = \max\{v_1, v_2\}$ follows a phase-type distribution $(\boldsymbol{\alpha}, \mathbf{Q}_{FJ})$ [32, 42, 44] defined on a four-state Markov process with infinitesimal generator matrix $\mathbf{Q}_{FJ}$ is expressed as

$$\mathbf{Q}_{FJ} = \begin{bmatrix} \tilde{\mathbf{Q}} & \tilde{\mathbf{q}} \\ \mathbf{0} & 0 \end{bmatrix}, \quad \tilde{\mathbf{Q}} = \begin{bmatrix} -(\hat{\mu}_1 + \hat{\mu}_2) & \hat{\mu}_1 & \hat{\mu}_2 \\ 0 & -\hat{\mu}_2 & 0 \\ 0 & 0 & -\hat{\mu}_1 \end{bmatrix}, \quad \tilde{\mathbf{q}} = \begin{bmatrix} 0 \\ \hat{\mu}_2 \\ \hat{\mu}_1 \end{bmatrix}, \quad (46)$$

together with the initial probability vector $(\boldsymbol{\alpha}, \alpha_{00})$ where $\boldsymbol{\alpha} = [1, 0, 0]$ and $\alpha_{00} = 0$. Defining $\exp(\mathbf{Y}) := \sum_{j=0}^{\infty} \frac{1}{j!} \mathbf{Y}^j$ for any given matrix $\mathbf{Y}$, the distribution function $H(\cdot)$ of the random variable $\psi(\emptyset)$ is given by

$$H(y) = 1 - \boldsymbol{\alpha} \exp(\tilde{\mathbf{Q}}y) \mathbf{e}, \quad y \in \mathbb{R}_{\geq 0},$$

from which we have $\mathbb{E}[\psi(\emptyset)] = -\boldsymbol{\alpha} \tilde{\mathbf{Q}}^{-1} \mathbf{e}$. Then we obtain from (46) and direct calculation

$$\tilde{\mathbf{Q}}^{-1} = \begin{bmatrix} -\frac{1}{\hat{\mu}_1 + \hat{\mu}_2} & -\frac{\hat{\mu}_1}{\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} & -\frac{\hat{\mu}_2}{\hat{\mu}_1(\hat{\mu}_1 + \hat{\mu}_2)} \\ 0 & -\frac{1}{\hat{\mu}_2} & 0 \\ 0 & 0 & -\frac{1}{\hat{\mu}_1} \end{bmatrix},$$

which together with $-\boldsymbol{\alpha} \tilde{\mathbf{Q}}^{-1} \mathbf{e}$ yields the desired result (8).

6.4 Proof of Lemma 3

We have by definition that

$$\psi(\{t\}) = \begin{cases} \max\{v_1, v_2\}, & t \in [0, \infty), \\ \max\{v_1 + (w_1 + t)^+, v_2 + (w_2 + t)^+\}, & t \in (-\infty, 0), \end{cases}$$

where $(x)^+ := \max\{0, x\}$, from which together with (10) we conclude

$$f^{(1)}(0) = \int_{-\infty}^0 (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt, \quad (47)$$

given that $\mathbb{E}[\psi(\{t\})] = \mathbb{E}[\max\{v_1, v_2\}] = \mathbb{E}[\psi(\emptyset)]$ for $t \in [0, \infty)$. By definition, the random variables $v_1 + (w_1 + t)^+$ and $v_2 + (w_2 + t)^+$ for fixed $t \in (-\infty, 0)$ are independent and respectively follow the distribution function given by

$$\begin{aligned} H_1(y) &= 1 - \hat{H}_1(y), & H_2(y) &= 1 - \hat{H}_2(y), & y &\in \mathbb{R}_{\geq 0}, \\ \hat{H}_1(y) &= (1 + \hat{\mu}_1 y \exp(\hat{\mu}_1 t)) \exp(-\hat{\mu}_1 y), & \hat{H}_2(y) &= (1 + \hat{\mu}_2 y \exp(\hat{\mu}_2 t)) \exp(-\hat{\mu}_2 y). \end{aligned}$$

From the independence of these two random variables, we obtain for fixed $t \in (-\infty, 0)$

$$\begin{aligned}
\mathbb{E}[\psi(\{t\})] &= \int_0^\infty (1 - H_1(y)H_2(y))dy \\
&= \int_0^\infty \hat{H}_1(y)dy + \int_0^\infty \hat{H}_2(y)dy - \int_0^\infty \hat{H}_1(y)\hat{H}_2(y)dy, \\
&= \frac{1 + e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{1 + e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3}. \tag{48}
\end{aligned}$$

Upon substituting (48) and (8) from Lemma 2 into (47), we then derive

$$\begin{aligned}
f^{(1)}(0) &= \int_{-\infty}^0 \left(\frac{e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3} + \frac{1}{\hat{\mu}_1} + \frac{1}{\hat{\mu}_2} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)} \right) dt \\
&= \int_{-\infty}^0 \left(\frac{e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3} \right) dt, \\
&= \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4},
\end{aligned}$$

thus yielding the desired result.

6.5 Proof of Lemma 4

Taking the limit as $r \rightarrow 1$, the application workload of the fork-join queueing network becomes balanced. The heavy-traffic limit for the fork-join queueing network with homogeneous service rates is known to be $h = 11/8$ [61], thus establishing the upper bound in (13). In the other direction, taking the limit as $r \rightarrow \infty$, the application workload of the fork-join queueing network becomes unbalanced to the point of reducing the stochastic network to the corresponding M/M/1 queue. The heavy-traffic limit for the M/M/1 queue is known to be $h = 1$ [22, 64], thus establishing the lower bound in (13).

Our upper-lower bound approximation smoothly interpolates between the upper-bound (homogeneous) limit of $11/8$ and the lower-bound (bottleneck) limit of 1 according to the power-law function in (14) with respect to the heterogeneity ratio r . The interpolation approximation satisfies $h(1) = 11/8$ for all values of β , thus matching the exact result for $r = 1$. The interpolation approximation further satisfies the lower bound $h(r) \rightarrow 1$ as $r \rightarrow \infty$, with β controlling the rate of decay from the upper bound toward the lower bound with respect to the heterogeneity ratio r .

6.6 Proof of Theorem 2

We start with the 0^{th} order light-heavy traffic interpolation approximation $\tilde{f}_0(\gamma)$ in (15) for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1]$, based on a combination of the 0^{th} order derivative and the heavy-traffic limit. The combination of (54), (56), and (57) in Appendix A yields

$$g(\gamma) = (\hat{\mu}_1 - \gamma) (f^{(0)}(0) + \gamma h),$$

from which together with (8) in Lemma 2 and (14) in Lemma 4 we derive

$$g(0) = (\hat{\mu}_1 - 0) \left(\frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} + 0 \cdot f^{(1)}(0) + 0 \cdot h \right) = \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)},$$

$$g(\mu^*) = g(\hat{\mu}_1) = h^* = h(r).$$

With respect to (55) in Appendix A, the linear interpolation $\tilde{g}(\gamma)$ corresponding to $g(\gamma)$ is given by

$$\tilde{g}(\gamma) = a_1\gamma + a_0.$$

From the matching light-traffic conditions (58) and the heavy-traffic condition (59) in Appendix A we have

$$a_1 = \frac{1}{\hat{\mu}_1}(h(r) - a_0) = \frac{1}{\hat{\mu}_1}h(r) - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}$$

$$a_0 = \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)},$$

from which we obtain

$$\begin{aligned} \tilde{g}(\gamma) &= \gamma \left(\frac{1}{\hat{\mu}_1}h(r) - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \right) + \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\ &= \frac{\gamma}{\hat{\mu}_1}h(r) + (\hat{\mu}_1 - \gamma) \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \end{aligned}$$

Then, given this linear interpolation $\tilde{g}(\gamma)$, we obtain from (60) in Appendix A the desired 0^{th} order interpolation approximation $\tilde{f}_0(\gamma)$ of $f(\gamma)$ as

$$\tilde{f}_0(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \frac{h(r)}{\hat{\mu}_1} + \frac{\hat{\mu}_1 - \gamma}{\hat{\mu}_1 - \gamma} \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \quad (49)$$

Next, we turn to the 1^{st} order light-heavy traffic interpolation approximation $\tilde{f}_1(\gamma)$ in (16) for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1)$, based on a combination of the 0^{th} order and 1^{st} order derivatives and the heavy-traffic limit. The combination of (54), (56), and (57) in Appendix A yields

$$g(\gamma) = (\hat{\mu}_1 - \gamma) \left(f^{(0)}(0) + \gamma f^{(1)}(0) + \frac{\gamma^2}{2}h \right),$$

from which together with (8) in Lemma 2, (11) in Lemma 3, and (14) in Lemma 4 we derive

$$\begin{aligned}
g(0) &= (\hat{\mu}_1 - 0) \left(\frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} + 0 \cdot f^{(1)}(0) + 0 \cdot h \right) = \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
g'(0) &= \frac{d}{d\gamma} \left[(\hat{\mu}_1 - \gamma) \left(f^{(0)}(0) + \gamma f^{(1)}(0) + \frac{\gamma^2}{2} h \right) \right] \Big|_{\gamma=0}, \\
&= \frac{d}{d\gamma} \left[\hat{\mu}_1 f^{(0)}(0) + \gamma \hat{\mu}_1 f^{(1)}(0) + \frac{\gamma^2 \hat{\mu}_1}{2} h - \gamma f^{(0)}(0) - \gamma^2 f^{(1)}(0) - \frac{\gamma^3}{2} h \right] \Big|_{\gamma=0}, \\
&= \left[0 + \hat{\mu}_1 f^{(1)}(0) + \gamma \hat{\mu}_1 h - f^{(0)}(0) - 2\gamma f^{(1)}(0) - \frac{3\gamma^2}{2} h \right] \Big|_{\gamma=0}, \\
&= \hat{\mu}_1 f^{(1)}(0) - f^{(0)}(0) = \frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
g(\mu^*) &= g(\hat{\mu}_1) = h^*.
\end{aligned}$$

With respect to (55) in Appendix A, the quadratic interpolation $\tilde{g}(\gamma)$ corresponding to $g(\gamma)$ is given by

$$\tilde{g}(\gamma) = a_2\gamma^2 + a_1\gamma + a_0.$$

From the matching light-traffic conditions (58) and the heavy-traffic condition (59) in Appendix A we have

$$\begin{aligned}
a_2 &= 0, \\
a_1 &= \frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
a_0 &= \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)},
\end{aligned}$$

from which we obtain

$$\begin{aligned}
\tilde{g}(\gamma) &= \gamma \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \right) + \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
&= \gamma \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + (\hat{\mu}_1 - \gamma) \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}
\end{aligned}$$

Then, given this quadratic interpolation $\tilde{g}(\gamma)$, we obtain from (60) in Appendix A the desired 1st order interpolation approximation $\tilde{f}_1(\gamma)$ of $f(\gamma)$ as

$$\tilde{f}_1(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1 - \gamma}{\hat{\mu}_1 - \gamma} \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \quad (50)$$

for $\gamma \in [0, \hat{\mu}_1)$, thus completing the proof.

6.7 Proof of Theorem 5

The Lagrangian associated with (RCA-OPT:FF) is defined as

$$\sum_{i=1}^N \frac{\omega_i \zeta_i}{\hat{\mu}_i - \gamma_i} + \chi \left(\sum_{i=1}^N c_i \hat{\mu}_i - \mathcal{C} \right), \quad (51)$$

where χ is the Lagrange multiplier. Taking the derivative of (51) with respect to $\hat{\mu}_i$ yields

$$\chi c_i - \frac{\omega_i \zeta_i}{(\hat{\mu}_i - \gamma_i)^2},$$

from which upon setting the result to zero and solving for $\hat{\mu}_i$ we obtain

$$\hat{\mu}_i = \gamma_i + \frac{1}{\sqrt{\chi}} \sqrt{\frac{\omega_i \zeta_i}{c_i}}. \quad (52)$$

Substituting (52) into (41) set to equality then renders

$$\sum_{i=1}^N c_i \hat{\mu}_i = \sum_{i=1}^N \gamma_i c_i + \frac{1}{\sqrt{\chi}} \sum_{i=1}^N \sqrt{\omega_i \zeta_i c_i} = \mathcal{C},$$

from which we have

$$\frac{1}{\sqrt{\chi}} = \frac{\mathcal{C} - \sum_{i=1}^N \gamma_i c_i}{\sum_{i=1}^N \sqrt{\omega_i \zeta_i c_i}}. \quad (53)$$

Upon substituting (53) into (52), we obtain the desired result

$$\hat{\mu}_i^* = \gamma_i + \left(\mathcal{C} - \sum_{j=1}^N \gamma_j c_j \right) \frac{\sqrt{\omega_i \zeta_i / c_i}}{\sum_{j=1}^N \sqrt{\omega_j \zeta_j c_j}}.$$

7 Numerical Experiments

In this section we present results from a large collection of numerical experiments that support and validate our general theoretical foundation for the mathematical optimization and performance analysis of resource allocation in quantum-centric supercomputing environments. We first consider numerical experiments for the two basic core components of quantum-centric supercomputing application workloads, starting with sequential computational phases and then moving to parallel computational phases.

7.1 Core Sequential Computational Phases

Let us consider the expected sojourn time $\mathbb{E}[T_{PF}]$ from (2) for the basic core sequential computational phase as a function of the arrival rate λ and the heterogeneity ratio r .

TBA: figure with 4 plots, each of which plots the expected sojourn time from (2) as a function of the heterogeneity ratio $r = 1, 2, 4, 8$, where the 4 plots are for different arrival rates $\lambda = 0.4, 0.8, 0.9, 0.98$ with $\hat{\mu}_1 = \hat{\mu}_3 = 1$, $\hat{\mu}_2 = \hat{\mu}_2 = r$, and $p_o = 0.5$ (but this is up to you).

TBA: a simple discussion of the results and their implications, with additional details to go in Appendix C

7.2 Core Parallel Computational Phases

Let us next consider the expected sojourn time $\mathbb{E}[T_{FJ}]$ from (6), (15), and (16) for the basic core parallel computational phase as a function of the arrival rate λ and the heterogeneity ratio r .

We first examine the accuracy of the three approximation for the fork-join queueing network, each of which provides the exact result when $r = 1$.

TBA: table that compares the expected sojourn time from (6), (15), and (16) with simulation results for different heterogeneity ratios $r = 1, 2, 4, 8$ and different arrival rates $\gamma = \lambda = 0.4, 0.8, 0.9, 0.98$ with $\hat{\mu}_1 = 1$, $\hat{\mu}_2 = r$ (but this is up to you).

We next consider the expected sojourn time $\mathbb{E}[T_{FJ}]$ for the basic core parallel computational phase as a function of the arrival rate λ and the heterogeneity ratio r .

TBA: figure with 4 plots, each of which plots the expected sojourn time from (6), (15), or (16) – choose whatever one you think is best – as a function of the heterogeneity ratio $r = 1, 2, 4, 8$, where the 4 plots are for different arrival rates $\gamma = \lambda = 0.4, 0.8, 0.9, 0.98$ with $\hat{\mu}_1 = 1$, $\hat{\mu}_2 = r$ (but this is up to you).

TBA: a simple discussion of the results and their implications, with additional details to go in Appendix C

7.2.1 Accuracy of Approximations

We consider three closed-form approximation methods for the mean response time of a heterogeneous two-queue M/M/1 fork-join system with heterogeneity ratio $r = \mu_1/\mu_2$, where $\mu_1 \geq \mu_2$.

- Upper/Lower Bound Interpolation (*UL*): T_{UL} , interpolates between a known independent upper bound and a bottleneck lower bound, using the average server utilization as the interpolation weight.
- Light/Heavy Traffic Interpolation (*LH*): T_{LH} , is derived from the Reiman–Simon light-heavy traffic interpolation framework; it matches the exact light-traffic value and a heavy-traffic limit $h(r)$.
- Enhanced Light/Heavy Traffic (*LH^{enh}*): T_{LH}^{enh} uses an additional matching of the first light-traffic derivative, yielding a quadratic numerator that improves accuracy at high heterogeneity.

All three approximations recover the exact expression for the homogeneous case. The approximations were validated against discrete-event simulation of the two-queue fork-join system. Each scenario used 20 M jobs with a 500 K-job warmup period, repeated over 5 independent random seeds; the reported T_{sim} is the grand mean and the 95% confidence interval is computed via the t -distribution across seeds ($\text{df} = 4$). Numerical results are illustrated in table 1.

Table 1: Approximations T_{UL} , T_{LH} , and $T_{\text{LH}}^{\text{enh}}$ vs. simulation for the heterogeneous two-queue fork-join ($\mu_1 = 1$, $\beta = 10$). $T_{\text{sim}} \pm 95\%$ CI from 10 M jobs (seed 42, normal approximation). Errors are $(T_{\text{approx}} - T_{\text{sim}})/T_{\text{sim}} \times 100\%$.

ρ	r	Simulation	T_{UL}		T_{LH}		$T_{\text{LH}}^{\text{enh}}$	
		$T_{\text{sim}} \pm 95\% \text{ CI}$	T	Err	T	Err	T	Err
0.4	1	2.419 ± 0.001	2.417	−0.09%	2.417	−0.09%	2.417	−0.09%
	2	1.816 ± 0.001	1.824	+0.46%	1.834	+0.97%	1.825	+0.48%
	4	1.704 ± 0.001	1.704	+0.00%	1.717	+0.76%	1.705	+0.04%
	8	1.677 ± 0.001	1.676	−0.10%	1.681	+0.19%	1.676	−0.09%
0.8	1	7.026 ± 0.004	7.000	−0.37%	7.000	−0.37%	7.000	−0.37%
	2	5.109 ± 0.003	5.101	−0.15%	5.168	+1.16%	5.151	+0.81%
	4	5.042 ± 0.003	5.016	−0.52%	5.050	+0.15%	5.026	−0.33%
	8	5.032 ± 0.003	5.003	−0.56%	5.014	−0.35%	5.005	−0.53%
0.9	1	13.956 ± 0.007	13.875	−0.58%	13.875	−0.58%	13.875	−0.58%
	2	10.133 ± 0.006	10.063	−0.69%	10.170	+0.37%	10.150	+0.17%
	4	10.094 ± 0.006	10.009	−0.84%	10.050	−0.43%	10.023	−0.70%
	8	10.088 ± 0.006	10.002	−0.85%	10.014	−0.73%	10.004	−0.84%
0.95	1	27.830 ± 0.014	27.625	−0.74%	27.625	−0.74%	27.625	−0.74%
	2	20.260 ± 0.012	20.036	−1.11%	20.174	−0.43%	20.153	−0.53%
	4	20.239 ± 0.012	20.005	−1.16%	20.050	−0.93%	20.021	−1.07%
	8	20.236 ± 0.012	20.001	−1.16%	20.014	−1.10%	20.003	−1.15%

7.2.2 Experimental Results

Among the three approximations, T_{UL} achieves the best uniform accuracy across all loads and heterogeneity ratios, remains within the theoretical bounds by construction, and requires no tunable parameters. To characterize how its accuracy depends on both dimensions simultaneously, we sweep the heterogeneity ratio $r = \mu_2/\mu_1$ from 1 to 8 across four load levels chosen with a geometric spacing in the slack to saturation, $1 - \rho \in \{0.48, 0.24, 0.12, 0.06\}$,

i.e. $\rho \in \{0.52, 0.76, 0.88, 0.94\}$.

The heterogeneity grid uses 8 values $r \in \{1, 1.15, 1.3, 1.5, 2, 3, 5, 8\}$, spaced more densely near $r = 1$ where T transitions most steeply from the homogeneous value to the bottleneck asymptote. Each of the 32 simulation points uses 10 M jobs with a 500 K-job warmup and a single random seed; 95% confidence intervals are computed by the normal approximation.

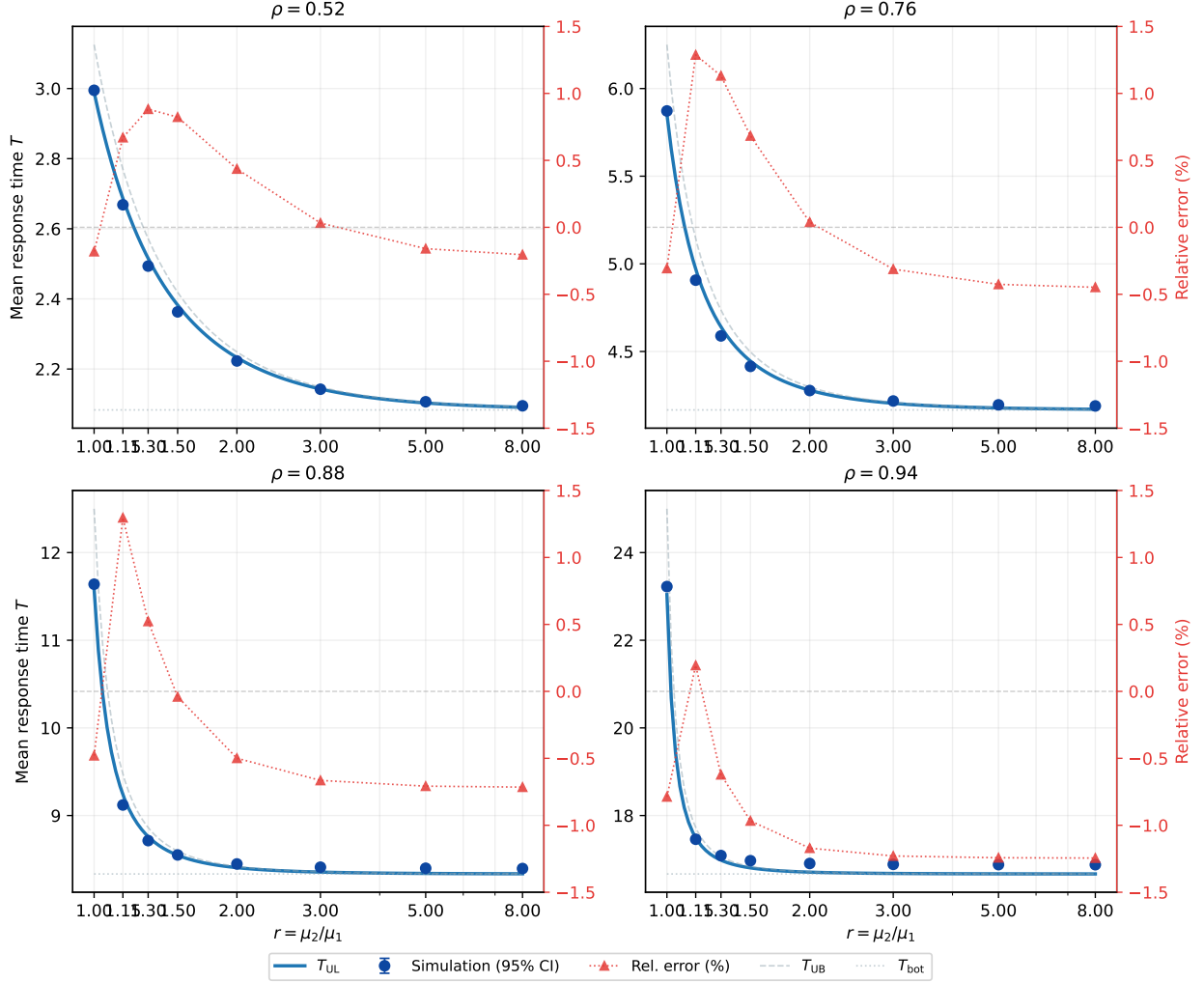


Figure 1: T_{UL} vs. heterogeneity ratio $r = \mu_{\max}/\mu_{\min}$ for four load levels $\rho \in \{0.52, 0.76, 0.88, 0.94\}$. Solid blue line: T_{UL} (analytical); filled circles: simulation with 95% CI bars (10 M jobs, seed = 42). Dashed and dotted gray lines: upper bound T_{UB} and lower bound T_{bot} . Right axis (red triangles): relative error $(T_{UL} - T_{sim})/T_{sim} \times 100\%$.

Figure 1 shows the results. At all four load levels, T_{UL} closely tracks the simulation. Most of the descent from the homogeneous value to the bottleneck plateau occurs in $r \in [1, 2]$; beyond $r \approx 3$ the curves are essentially flat. The relative error (right axis) peaks near $r = 1.15$ – 1.3 and is bounded below 1.3% for $\rho \leq 0.88$. At the highest load ($\rho = 0.94$), the

single-seed estimate at $r = 1$ lies slightly above the exact homogeneous value, a reminder that estimates near saturation require either multiple seeds or substantially longer runs to eliminate transient bias; the displayed CI underestimates the true uncertainty in this regime because consecutive response times are highly correlated near saturation.

7.3 Optimal Resource Capacity Allocation

Finally, let us turn to consider the optimal resource capacities from the solution of (RCA-OPT) together with the corresponding expected sojourn time $\mathbb{E}[T]$ as a function of the arrival rate λ , heterogeneity ratio r , weight vector ω , cost vector $\mathbf{c}$, and budgetary cost $\mathcal{C}$.

TBA: Plot results for optimal resource capacities $\hat{\mu}$ for different parameter values. see range of parameter values provided below as part of the optimization iteration.

TBA: Plot corresponding results for $\mathbb{E}[T_{FJ}]$ under the above optimal resource capacities $\hat{\mu}$ for different arrival rates λ , heterogeneous ratios r , cost budgets $\mathcal{C}$. Compare these results against the corresponding $\mathbb{E}[T_{FJ}]$ results under the resource allocation reservation policy of Section 4 with the combined optimal resource capacities $\hat{\mu}$

SECTION UNDER CONSTRUCTION

The queueing network of our quantum-centric supercomputing environment is illustrated in Figure 2. This represents an illustration of the model described in Section 2.

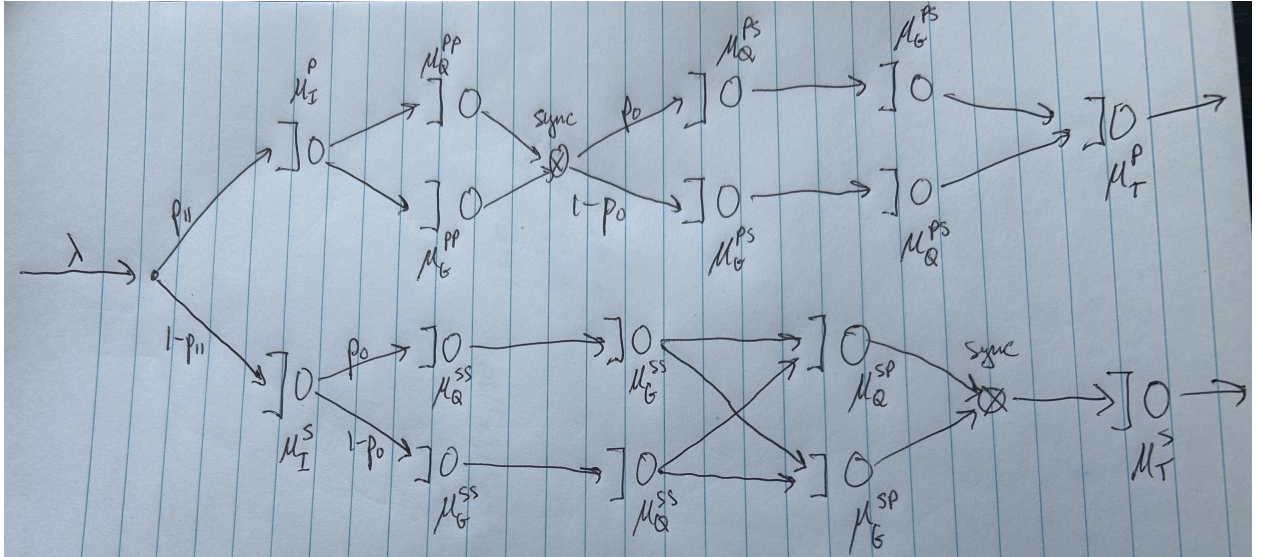


Figure 2: Illustration of Quantum-Centric Supercomputing Environment. This is solely intended to illustrate the queueing network that we want to simulate.

Given our assumption in Section 2 that $\mu_I^P, \mu_I^S, \mu_T^P, \mu_T^S \gg \mu_Q^{PP}, \mu_Q^{PS}, \mu_Q^{SP}, \mu_G^{PP}, \mu_G^{PS}, \mu_G^{SP}$, the service rates $\mu_I^P = \mu_I^S = \mu_T^P = \mu_T^S$ for the classical initialization and termination tasks on

the system CPUs are fixed to be very, very large relative to the service rates for the other tasks in the queueing network; i.e., the mean service times $1/\mu_I^P = 1/\mu_I^S = 1/\mu_T^P = 1/\mu_T^S$ for the classical initialization and termination tasks on the system CPUs are fixed to be very, very small relative to the service rates for the other tasks in the queueing network. Let us therefore say that $1/\mu_I^P = 1/\mu_I^S = 1/\mu_T^P = 1/\mu_T^S = 0.0001$.

Recall the heterogeneity ratio $r := \hat{\mu}_2/\hat{\mu}_1 \geq 1$. For quantum-dominant application workloads, we have $\mu_G^{PP} = r \cdot \mu_Q^{PP}$, $\mu_G^{PS} = r \cdot \mu_Q^{PS}$, and $\mu_G^{SP} = r \cdot \mu_Q^{SP}$. Let us then set $\mu_Q^{PP} = \mu_Q^{PS} = \mu_Q^{SP} = 1$, and thus $\mu_G^{PP} = r$, $\mu_G^{PS} = r$, and $\mu_G^{SP} = r$. Analogously, for classical-dominant application workloads, we have $\mu_Q^{PP} = r \cdot \mu_G^{PP}$, $\mu_Q^{PS} = r \cdot \mu_G^{PS}$, and $\mu_Q^{SP} = r \cdot \mu_G^{SP}$. Let us then set $\mu_G^{PP} = \mu_G^{PS} = \mu_G^{SP} = 1$, and thus $\mu_Q^{PP} = r$, $\mu_Q^{PS} = r$, and $\mu_Q^{SP} = r$. Our experiments will consist of scenarios in which we consider a wide range of values for $r \geq 1$. Given the symmetry of the queueing network structure, the performance and optimal solutions solely depend on the value of r and not on whether the application workload is quantum-dominant or classical-dominant.

Let us therefore say that we experiment with the scenarios $r = 1$, $r = 2$, $r = 4$, and $r = 8$. In each case, we need to make sure we consider a cost budget $\mathcal{C}$ that is sufficiently large to ensure $\hat{\mu}_i > \gamma_i$ for all queues i , where $i \in \{1, \dots, 16\}$ (see Figure 2). Then, we can consider three basic budgetary cost constraints: one in which the budget is modestly above the requirements for $\hat{\mu}_i > \gamma_i$, one in which the budget is much larger than this modest budgetary cost constraint, and one that is in between these two extremes.

Here are the details required for generating the results/plots for this section.

Let us number the queues in Figure 2 from left to right and top to bottom as $1, \dots, 16$. Then fix $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}_{15} = \hat{\mu}_{16} = 10,000$. For a given λ , set:

$$\begin{aligned}
\gamma_1 &= p_{\parallel}\lambda; \\
\gamma_2 &= (1 - p_{\parallel})\lambda; \\
\gamma_3 &= p_{\parallel}\lambda; \\
\gamma_4 &= p_{\parallel}\lambda; \\
\gamma_5 &= (1 - p_{\parallel})p_{\circ}\lambda; \\
\gamma_6 &= (1 - p_{\parallel})(1 - p_{\circ})\lambda; \\
\gamma_7 &= p_{\parallel}p_{\circ}\lambda; \\
\gamma_8 &= p_{\parallel}(1 - p_{\circ})\lambda; \\
\gamma_9 &= (1 - p_{\parallel})p_{\circ}\lambda; \\
\gamma_{10} &= (1 - p_{\parallel})(1 - p_{\circ})\lambda; \\
\gamma_{11} &= p_{\parallel}p_{\circ}\lambda; \\
\gamma_{12} &= p_{\parallel}(1 - p_{\circ})\lambda; \\
\gamma_{13} &= (1 - p_{\parallel})\lambda; \\
\gamma_{14} &= (1 - p_{\parallel})\lambda; \\
\gamma_{15} &= p_{\parallel}\lambda; \\
\gamma_{16} &= (1 - p_{\parallel})\lambda.
\end{aligned}$$

Consider the following set of parameters:

$\lambda = 0.4, 0.8, 0.9, 0.98$ (or some variation thereof);

$r = 1, 2, 4, 8$;

$p_{\parallel} = 0.2, 0.5, 0.8$ (or some variation thereof);

$p_{\circ} = 0.2, 0.5, 0.8$ (or some variation thereof);

Two cost scenarios (all equal, quantum 4 times more):

(1) $c_3, c_5, c_7, c_{10}, c_{12}, c_{13} = c_4, c_6, c_8, c_9, c_{11}, c_{14} = 1$

(2) $c_3, c_5, c_7, c_{10}, c_{12}, c_{13} = 4$ and $c_4, c_6, c_8, c_9, c_{11}, c_{14} = 1$ (or some variation thereof)

It is not clear what values to consider for the cost budget $\mathcal{C}$, so we will have to experiment with this.

Then, for a given set of parameter values for λ , r , $p_{\parallel}$, $p_{\circ}$ and $\mathcal{C}$, we compute the following loop:

- Set $\zeta_i^{(0)} = 1$ for all $i = 3, 4, \dots, 13, 14$
- For $j = 1, 2, \dots$:
 - Compute $\hat{\mu}_i^{(j)}$ from $\zeta_i^{(j-1)}$ using equation (43) for all $i = 3, 4, \dots, 13, 14$
 - Simulate network of Figure 2 with input $\lambda, \hat{\mu}_i^{(j)}, r, p_{\parallel}, p_{\circ}, c_i$, and returning output $\mathbb{E}[T_i^{(j)}]$ for all $i = 3, \dots, 14$ where $\mathbb{E}[T_i^{(j)}]$ is the mean sojourn time for queue i
 - Compute $\zeta_i^{(j)} = \mathbb{E}[T_i^{(j)}](\hat{\mu}_i^{(j)} - \gamma_i)$ following equation (44) for all $i = 3, \dots, 14$
 - If $(\hat{\mu}_i^{(j)} - \hat{\mu}_i^{(j-1)}) < \epsilon$ for all $i = 3, \dots, 14$, then BREAK
- Return $\mathbb{E}[T_i^{(j)}]$ and $\hat{\mu}_i^{(j)}$ for all $i = 3, \dots, 14$

8 Conclusions

TBA

References

- [1] A. Ajagekar, K. A. Hamoud, and F. You. Hybrid classical-quantum optimization techniques for solving mixed-integer programming problems in production scheduling. *IEEE Transactions on Quantum Engineering*, 3:1–16, 2022.
- [2] S. Asmussen. *Applied Probability and Queues*. Springer, second edition, 2003.
- [3] S. Asmussen and P. W. Glynn. *Stochastic Simulation: Algorithms and Analysis*. Springer, 2007.

- [4] F. Baccelli, A. M. Makowski, and A. Shwartz. The fork-join queue and related systems with synchronization constraints: Stochastic ordering and computable bounds. *Advances in Applied Probability*, 21(3):629–660, 1989.
- [5] S. Balsamo, L. Donatiello, and N. Van Dijk. Bound performance models of heterogeneous parallel processing systems. *IEEE Transactions on Parallel and Distributed Systems*, 9(10):1041–1056, 1998.
- [6] F. Baskett, K. M. Chandy, R. R. Muntz, and F. G. Palacios. Open, closed, and mixed networks of queues with different classes of customers. *Journal of the Association for Computing Machinery*, 22(2):248–260, 1975.
- [7] N. G. Bean, J.-M. Li, and P. G. Taylor. Caudal characteristics of qbds with decomposable phase spaces. In G. Latouche and P. Taylor, editors, *Advances in Algorithmic Methods for Stochastic Models*, pages 37–55. Notable Publications, 2000.
- [8] T. Beck, A. Baroni, R. Bennink, G. Buchs, E. A. C. Pérez, M. Eisenbach, R. F. da Silva, M. G. Meena, K. Gottiparthi, P. Groszkowski, T. S. Humble, R. Landfield, K. Maheshwari, S. Oral, M. A. Sandoval, A. Shehata, I.-S. Suh, and C. Zimmer. Integrating quantum computing resources into scientific hpc ecosystems. *Future Generation Computer Systems*, 161:11–25, 2024.
- [9] P. Benioff. The computer as a physical system: A microscopic quantum mechanical hamiltonian model of computers as represented by turing machines. *Journal of Statistical Physics*, 22(5):563–591, 1980.
- [10] G. Brassard, P. Høyer, M. Mosca, and A. Tapp. Quantum amplitude amplification and estimation. *Contemporary Mathematics*, 305:53–74, 2002.
- [11] P. Brémaud. *Markov Chains: Gibbs Fields, Monte Carlo Simulation and Queues*. Springer, second edition, 2020.
- [12] P. J. Burke. The output of a queueing system. *Operations Research*, 4(6):699–704, 1956.
- [13] H. Chen and D. D. Yao. *Fundamentals of Queueing Networks: Performance, Asymptotics, and Optimization*. Springer, 2001.
- [14] A. B. Dieker, S. Ghosh, and M. S. Squillante. Optimal resource capacity management for stochastic networks. *Operations Research*, 65(1):221–241, 2017.
- [15] R. P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6/7):467–488, 1982.
- [16] L. Flatto. Two parallel queues created by arrivals with two demands II. *SIAM Journal of Applied Mathematics*, 45:861–878, 1985.

- [17] L. Flatto and S. Hahn. Two parallel queues created by arrivals with two demands I. *SIAM Journal of Applied Mathematics*, 44:1041–1053, 1984.
- [18] D. Georg, J. Barzen, M. Beisel, F. Leymann, J. Obst, D. Vietz, B. Weder, and V. Yusupov. Execution patterns for quantum applications. In *ICSOF*T, pages 258–268, 2023.
- [19] T. Goda, T. Hironaka, W. Kitade, and A. Foster. Unbiased mlmc stochastic gradient-based optimization of bayesian experimental designs. *SIAM Journal on Scientific Computing*, 44(1):A286–A311, 2022.
- [20] L. K. Grover. A fast quantum mechanical algorithm for database search. In *ACM Symposium on Theory of Computing*, pages 212–219. ACM, 1996.
- [21] L. K. Grover. Quantum mechanics helps in searching for a needle in a haystack. *Physical Review Letters*, 79(2):325–328, 1997.
- [22] J. M. Harrison. *Brownian Motion and Stochastic Flow Systems*. J. Wiley and Sons, 1985.
- [23] A. W. Harrow, A. Hassidim, and S. Lloyd. Quantum algorithm for linear systems of equations. *Physical Review Letters*, 103(15):150502, 2009.
- [24] Y. Hu, J. Wang, Y. Xie, A. Krause, and D. Kuhn. Contextual stochastic bilevel optimization. *Advances in Neural Information Processing Systems*, 36, 2024.
- [25] V. Istratescu. *Fixed Point Theory: An Introduction*. Volume 7 of Mathematics and Its Applications. Springer, 2001.
- [26] J. R. Jackson. Jobshop-like queueing systems. *Management Science*, 10(1):131–142, 1963.
- [27] F. P. Kelly. *Reversibility and Stochastic Networks*. Wiley, 1979.
- [28] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, and A. Kandala. Evidence for the utility of quantum computing before fault tolerance. *Nature*, 618(7965):500–505, 2023.
- [29] L. Kleinrock. *Message Delay in Communication Nets with Storage*. PhD thesis, Massachusetts Institute of Technology, 1963.
- [30] R. Kothari and R. O’Donnell. Mean estimation when you have the source code; or, quantum monte carlo methods. In *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1186–1215. SIAM, 2023.
- [31] H. J. Kushner and G. G. Yin. *Stochastic Approximation and Recursive Algorithms and Applications*. Springer, second edition, 2003.

- [32] G. Latouche and V. Ramaswami. *Introduction to Matrix Analytic Methods in Stochastic Modeling*. ASA-SIAM, Philadelphia, 1999.
- [33] S. S. Lavenberg, editor. *Computer Performance Modeling Handbook*. Academic Press, 1983.
- [34] D. A. Levin, Y. Peres, and E. L. Wilmer. *Markov Chains and Mixing Times*. American Mathematical Society, second edition, 2017.
- [35] N. Ma and H. Li. Understanding and estimating the execution time of quantum circuits, 2025.
- [36] P. Mantha, F. J. Kiwit, N. Saurabh, S. Jha, and A. Luckow. Pilot-Quantum: A Quantum-HPC middleware for resource, workload and task management, 2025.
- [37] M. Mohanty, G. Gautam, V. Aggarwal, and P. Parag. Analysis of fork-join scheduling on heterogeneous parallel servers. *IEEE/ACM Transactions on Networking*, 32(6):4798–4809, 2024.
- [38] A. Montanaro. Quantum speedup of monte carlo methods. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 471(2181), 2015.
- [39] R. R. Muntz. Poisson departure process and queueing networks. Technical Report RC 4145, IBM Research, 1972.
- [40] B. L. Nelson and S. G. Henderson, editors. *Handbooks in OR and MS: Simulation*. North-Holland, 2007.
- [41] R. D. Nelson and A. N. Tantawi. Approximate analysis of fork/join synchronization in parallel queues. *IEEE Transactions on Computers*, 37(6):739–743, June 1988.
- [42] M. F. Neuts. Probability distributions of phase type. In *Liber Amicorum Prof. Emeritus H. Florin*, Department of Mathematics, University of Louvain, Belgium, pages 173–206, 1975.
- [43] M. F. Neuts. *Matrix-Geometric Solutions in Stochastic Models: An Algorithmic Approach*. 1981.
- [44] M. F. Neuts. *Matrix-Geometric Solutions in Stochastic Models: An Algorithmic Approach*. The Johns Hopkins University Press, 1981.
- [45] M. F. Neuts. The caudal characteristic curve of queues. *Advances in Applied Probability*, 18:221–254, 1986.
- [46] M. F. Neuts. *Structured Stochastic Matrices of M/G/1 Type and Their Applications*. Marcel Dekker, 1989.

- [47] V. Nguyen. The trouble with diversity: Fork-join networks with heterogeneous customer population. *Ann. Appl. Probab.*, 4(4):1–25, 1994.
- [48] P. Rebentrost, B. Gupt, and T. R. Bromley. Quantum computational finance: Monte carlo pricing of financial derivatives. *Physical Review A*, 98(2):022321, 2018.
- [49] M. I. Reiman and B. Simon. An interpolation approximation for queueing systems with poisson input. *Operations Research*, 36(3):454–469, 1988.
- [50] M. I. Reiman and B. Simon. Light traffic limits of sojourn time distributions in markovian queueing networks. *Stochastic Models*, 4(2):191–233, 1988.
- [51] M. I. Reiman and B. Simon. Open queueing systems in light traffic. *Mathematics of Operations Research*, 14(1):26–59, February 1989.
- [52] I. Rončević, F. Paschke, Y. Gao, L.-A. Lieske, L. A. Gödde, S. Barison, S. Piccinelli, A. Baiardi, I. Tavernelli, J. Repp, F. Albrecht, H. L. Anderson, and L. Gross. A molecule with half-Möbius topology. *Science*, 2026.
- [53] D. Schol, M. Vlasiou, and B. Zwart. Extreme values for the waiting time in large fork-join queues. *Queueing Systems*, 2025.
- [54] S. Seelam, J. M. Chow, A. Corcoles, S. Sheldon, T. Mittal, A. Kandala, S. Dague, I. Hincks, H. Horii, B. Johnson, M. Le, H. Jamjoom, and J. M. Gambetta. Reference architecture of a quantum-centric supercomputer, 2026.
- [55] E. Seneta. *Non-Negative Matrices and Markov Chains*. Springer Verlag, second edition, 1981.
- [56] A. Shehata, P. Groszkowski, T. Naughton, M. G. Meena, E. Wong, D. Claudino, R. F. da Silva, and T. Beck. Bridging paradigms: Designing for HPC-Quantum convergence. *Future Generation Computer Systems*, 174:107980, 2026.
- [57] P. W. Shor. Algorithms for quantum computation: discrete logarithms and factoring. In *Proceedings 35th Annual Symposium on Foundations of Computer Science*, pages 124–134, 1994.
- [58] P. W. Shor. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*, 26(5):1484–1509, 1997.
- [59] M. Slys, P. Rydlichowski, K. Kurowski, O. Bacarreza, E. C. Gomez, Z. Chandani, B. Heim, P. Khalate, W. R. Clements, and J. Fletcher. Hybrid Classical-Quantum supercomputing: A demonstration of a multi-user, multi-QPU and multi-GPU environment, 2025.
- [60] S. Varma. *Heavy and light traffic approximations for queues with synchronization constraints*. PhD thesis, University of Maryland, College Park, MD, USA, 1990.

- [61] S. Varma and A. M. Makowski. Interpolation approximations for symmetric fork-join queues. *Performance Evaluation*, 20:245–265, 1994.
- [62] B. Weder, U. Breitenbücher, F. Leymann, and K. Wild. Integrating quantum computing into workflow modeling and execution. In *2020 IEEE/ACM 13th International Conference on Utility and Cloud Computing (UCC)*, pages 279–291, 2020.
- [63] L. M. Wein. Capacity allocation in generalized Jackson networks. *Operations Research Letters*, 8:143–146, 1989.
- [64] W. Whitt. *Stochastic-Process Limits: An Introduction to Stochastic-Process Limits and Their Application to Queues*. Springer, 2002.
- [65] S. Woerner and D. J. Egger. Quantum risk analysis. *npj Quantum Information*, 5:1098–1105, 2019.
- [66] R. W. Wolff. *Stochastic Modeling and the Theory of Queues*. Prentice Hall, 1989.

A Light-Heavy Traffic Interpolation Approximation

In this appendix, we provide a general overview of the light-heavy traffic interpolation approximation approach which includes within this context the theoretical results of Section 3.2.2. The light-heavy traffic interpolation approximation can be an effective method for analyzing otherwise intractable queueing systems. The central idea is to approximate some quantity $f(\gamma)$ of interest in the system based on a combination of n light-traffic derivatives associated with the Taylor series expansion of $f(\gamma)$ at $\gamma = 0$ and the heavy-traffic limit of $f(\gamma)$ as γ approaches its critical value μ^* . More specifically, following the approach developed by Reiman and Simon [49–51], we first scale $f(\gamma)$ by defining

$$g(\gamma) := (\mu^* - \gamma)f(\gamma), \quad \gamma \in [0, \mu^*), \quad (54)$$

and then we interpolate $g(\gamma)$ by a polynomial of order $n + 1$ of the form

$$\tilde{g}(\gamma) = a_{n+1}\gamma^{n+1} + a_n\gamma^n + \dots + a_1\gamma + a_0, \quad (55)$$

where $a_{n+1}, a_n, \dots, a_1, a_0$ are determined from the n light-traffic derivatives associated with the Taylor series expansion of $f(\gamma)$ at $\gamma = 0$, i.e.,

$$f(\gamma) \approx f^{(0)}(0) + \gamma f^{(1)}(0) + \dots + \frac{\gamma^{n-1}}{(n-1)!} f^{(n-1)}(0) + \frac{\gamma^n}{n!} f^{(n)}(0), \quad \gamma \approx 0, \quad (56)$$

together with the heavy-traffic limit

$$\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma)f(\gamma) = h. \quad (57)$$

Here, assuming the existence of the first n derivatives of $f(\gamma)$ at $\gamma = 0$, the $f^{(k)}(0)$ in (56) denotes the order k derivative of $f(\gamma)$ at $\gamma = 0$, namely

$$f^{(k)}(0) = \left. \frac{d^k f}{d\gamma^k} \right|_{\gamma=0}, \quad k \in \{0, 1, \dots, n\}.$$

From (56) and (57), we have the matching light-traffic conditions

$$\tilde{g}^{(k)}(0) = g^{(k)}(0), \quad k \in \{0, 1, \dots, n\} \quad (58)$$

and the heavy-traffic condition

$$\tilde{g}(\mu^*) = g(\mu^*) = h. \quad (59)$$

Then, once $\tilde{g}(\gamma)$ is determined, we invert the scaling in (54) to obtain the interpolation approximation of $f(\gamma)$ as

$$\tilde{f}(\gamma) = \frac{\tilde{g}(\gamma)}{\mu^* - \gamma}, \quad \gamma \in [0, \mu^*). \quad (60)$$

The calculation of the derivatives at zero for the quantity of interest $f(\gamma)$ is clearly a focal point of the light-heavy traffic interpolation approximation approach. This is made possible within the context of Poisson (or Poisson-driven) arrival processes by the light-traffic theory rigorously developed in [51], where the authors furnish expressions that, in principle, enable us to calculate the derivative of any order n . The availability of the light traffic derivatives $f^{(k)}(0)$ for all k fully determines $f(\gamma)$ in terms of its Taylor series expansion, and thus provides a way to obtain an exact expression $\tilde{f}(\gamma) = f(\gamma)$. However, in our particular situation as well as in most practical cases, the calculations quickly become extremely involved and we therefore have to restrict ourselves to the 0^{th} and 1^{st} order derivatives. Nevertheless, we are strongly encouraged by the fact that our light-heavy traffic interpolation approximation for the complex unbalanced application workload with nonhomogeneous service rates $\hat{\mu}_1$ and $\hat{\mu}_2$ recovers the exact expression (3) for the expected sojourn time under the balanced workload with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$ due to Nelson and Tantawi [41]. Moreover, an a posteriori comparison of the resulting approximations with simulations strongly suggests that our quadratic polynomial interpolation in γ is an effective approximation.

Our interests are in deriving a light-heavy traffic interpolation approximation for the more difficult problem of approximating the expected sojourn time of jobs in a fork-join queueing network comprising two parallel servers under unbalanced application workloads in which the arrivals follow a Poisson process with rate γ and the nonhomogeneous service times for each of the two servers respectively follow an exponential distribution with rates $\hat{\mu}_1$ and $\hat{\mu}_2$. The quantity of interest in the system is therefore $f(\gamma) = \mathbb{E}[T_{FJ}]$ whose approximation $\tilde{f}(\gamma) = \mathbb{E}[\tilde{T}_{FJ}]$ is based on a combination of the 0^{th} and 1^{st} order derivatives $f^{(0)}(0)$ and $f^{(1)}(0)$ associated with the Taylor series expansion of $f(\gamma)$ at $\gamma = 0$ in (56) and the heavy-traffic limit of $f(\gamma)$ as γ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$ in (57). We note that the fork-join queueing network of interest satisfies the relevant technical conditions required by the light-heavy traffic interpolation approximation approach of Reiman and Simon [49–51, 60].

Following [49–51], consider the sojourn time T_{FJ} of a tagged job that arrives at time $t = 0$ whose service times v_1 and v_2 are random variables that follow an exponential distribution with mean $\mathbb{E}[v_1] = \hat{\mu}_1$ and $\mathbb{E}[v_2] = \hat{\mu}_2$, respectively, where v_1 and v_2 are independent of each other as well as all other random variables of the fork-join queueing network. The 0^{th} order derivative can be obtained by definition as

$$f^{(0)}(0) = \mathbb{E}[\psi(\emptyset)] := \mathbb{E}[T_{FJ} \mid \text{no arrivals over } (-\infty, 0)], \quad (7)$$

noting that any arrival at time $t \in (0, \infty)$ will be serviced after the tagged job and therefore not impact T_{FJ} . We then have the following theoretical result for $f^{(0)}(0)$.

Lemma 2. *The 0^{th} order derivative of the fork-join queueing network is given by*

$$f^{(0)}(0) = \mathbb{E}[\psi(\emptyset)] = \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \quad (8)$$

For the 1^{st} order derivative, consider the sojourn time T_{FJ} of a tagged job that arrives at time 0 with exponential service times v_1 and v_2 as above, given that another job arrives at time $t \in (-\infty, \infty)$ whose service times w_1 and w_2 are random variables that follow an exponential distribution with mean $\mathbb{E}[w_1] = \hat{\mu}_1$ and $\mathbb{E}[w_2] = \hat{\mu}_2$, respectively, where v_1, v_2, w_1, w_2 are independent of each other as well as all other random variables of the fork-join queueing network. The 1^{st} order derivative can be obtained by definition as

$$\mathbb{E}[\psi(\{t\})] := \mathbb{E}[T_{FJ} \mid \text{exactly one arrival at time } t \text{ over } (-\infty, \infty)], \quad (9)$$

$$f^{(1)}(0) = \int_{-\infty}^{\infty} (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt. \quad (10)$$

We then have the following theoretical result for $f^{(1)}(0)$.

Lemma 3. *The 1^{st} order derivative of the fork-join queueing network is given by*

$$f^{(1)}(0) = \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4}. \quad (11)$$

Next, we consider the heavy-traffic limit of $f(\gamma)$ in (57) as γ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. Define $r := \hat{\mu}_2/\hat{\mu}_1 \geq 1$. Without loss of generality, suppose that $\hat{\mu}_1 \leq \hat{\mu}_2$. Our approach consists of first establishing upper and lower bounds on the heavy-traffic limit h^* as $r \rightarrow 1$ and as $r \rightarrow \infty$, and then obtaining an upper-lower bound interpolation approximation for h^* . The corresponding theoretical results for the heavy-traffic limit h^* under any given $\hat{\mu}_1$ and $\hat{\mu}_2$ are presented as follows.

Lemma 4. *The heavy-traffic limit of the fork-join queueing network is bounded above and below by*

$$\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma) \mathbb{E}[T_{FJ}] = h^* \in [1, 11/8]. \quad (13)$$

Furthermore, the upper-lower bound interpolation approximation for this heavy-traffic limit is given by

$$h^* \approx h(r) = 1 + \frac{3}{8}r^{-\beta} \in [1, 11/8], \quad r \in [1, \infty), \quad \beta \in (0, \infty). \quad (14)$$

We now exploit the above results to obtain the desired 0^{th} and 1^{st} order light-heavy traffic interpolation approximations $\tilde{f}_0(\gamma)$ and $\tilde{f}_1(\gamma)$ for the expected sojourn time of jobs in the two-server fork-join queueing network of interest under unbalanced application workloads.

Theorem 2. *Consider the sojourn time T_{FJ} of jobs in a fork-join queueing network comprising two parallel servers where arrivals follow a Poisson process with rate γ and service times follow an exponential distribution with nonhomogeneous rates $\hat{\mu}_1$ and $\hat{\mu}_2$ at server 1 and 2, respectively. Further let $f(\gamma) = \mathbb{E}[T_{FJ}]$ and, without loss of generality, $\hat{\mu}_1 = \mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. The 0^{th} and 1^{st} order light-heavy traffic interpolation approximations $\tilde{f}_0(\gamma)$ and $\tilde{f}_1(\gamma)$ for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1)$, are then given by*

$$\tilde{f}_0(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \quad (15)$$

$$\tilde{f}_1(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \quad (16)$$

Remark 2. *It is important to note that, upon substituting $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}$ in (15) and (16), our light-heavy traffic interpolation approximations recover the exact expected sojourn time of jobs in the two-server fork-join queueing network of interest under balanced application workloads, as specifically expressed in (3).*

B Mathematical Analysis of Reserved Allocation

In this appendix, we provide some additional details on the matrix-analytic analysis of the quantum-centric supercomputing environment under the resource allocation reservation policy in Section 4.

The matrices corresponding to the non-homogeneous boundary of the state space $\hat{\Omega}$ are given by

$$\mathbf{B}_{00} = [-\lambda], \quad \mathbf{B}_{01} = [\mathbf{B}_{01}^P \quad \mathbf{B}_{01}^S], \quad \mathbf{B}_{10} = [\mathbf{B}_{10}^P \quad \mathbf{B}_{10}^S]^\top, \quad (18)$$

$$\mathbf{B}_{01}^P = [p_0\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad \mathbf{B}_{01}^S = [(1-p_0)\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad (19)$$

$$\mathbf{B}_{10}^P = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^P], \quad \mathbf{B}_{10}^S = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^S], \quad (20)$$

$$\mathbf{B}_{11} = \begin{bmatrix} \mathbf{B}_{11}^P & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{11}^S \end{bmatrix}, \quad \mathbf{B}_{11}^P = \begin{bmatrix} \mathbf{B}_{11}^{P_{pp}} & \mathbf{B}_{11}^{P_{ps}} \\ \mathbf{0} & \mathbf{B}_{11}^{P_{ss}} \end{bmatrix}, \quad \mathbf{B}_{11}^S = \begin{bmatrix} \mathbf{B}_{11}^{S_{pp}} & \mathbf{B}_{11}^{S_{ps}} \\ \mathbf{B}_{11}^{S_{sp}} & \mathbf{B}_{11}^{S_{ss}} \end{bmatrix}, \quad (21)$$

$$\mathbf{B}_{11}^{P_{pp}} = \begin{bmatrix} (\lambda, \mu_I^P) & 0 & 0 & \mu_I^P \\ 0 & (\lambda, \mu_G^{PP}) & 0 & 0 \\ 0 & 0 & (\lambda, \mu_Q^{PP}) & 0 \\ 0 & \mu_Q^{PP} & \mu_G^{PP} & (\lambda, \mu_Q^{PP}, \mu_G^{PP}) \end{bmatrix}, \quad (61)$$

$$\mathbf{B}_{11}^{P_{ps}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ (1-p_\circ)\mu_G^{PP} & 0 & 0 & p_\circ\mu_G^{PP} & 0 \\ (1-p_\circ)\mu_Q^{PP} & 0 & 0 & p_\circ\mu_Q^{PP} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (62)$$

$$\mathbf{B}_{11}^{P_{ss}} = \begin{bmatrix} (\lambda, \mu_G^{PS}) & \mu_G^{PS} & 0 & 0 & 0 \\ 0 & (\lambda, \mu_Q^{PS}) & 0 & 0 & \mu_Q^{PS} \\ 0 & 0 & (\lambda, \mu_G^{PS}) & 0 & \mu_G^{PS} \\ 0 & 0 & \mu_Q^{PS} & (\lambda, \mu_Q^{PS}) & 0 \\ 0 & 0 & 0 & 0 & (\lambda, \mu_T^P) \end{bmatrix}, \quad (63)$$

$$\mathbf{B}_{11}^{S_{pp}} = \begin{bmatrix} (\lambda, \mu_I^S) & 0 & 0 & 0 \\ 0 & (\lambda, \mu_G^{SP}) & 0 & 0 \\ 0 & 0 & (\lambda, \mu_Q^{SP}) & 0 \\ 0 & \mu_Q^{SP} & \mu_G^{SP} & (\lambda, \mu_G^{SP}, \mu_Q^{SP}) \end{bmatrix}, \quad (64)$$

$$\mathbf{B}_{11}^{S_{ps}} = \begin{bmatrix} (1-p_\circ)\mu_I^S & 0 & 0 & p_\circ\mu_I^S & 0 \\ 0 & 0 & 0 & 0 & \mu_G^{SP} \\ 0 & 0 & 0 & 0 & \mu_Q^{SP} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (65)$$

$$\mathbf{B}_{11}^{S_{sp}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu_Q^{SS} \\ 0 & 0 & 0 & \mu_G^{SS} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{11}^{S_{ss}} = \begin{bmatrix} (\lambda, \mu_G^{SS}) & \mu_G^{SS} & 0 & 0 & 0 \\ 0 & (\lambda, \mu_Q^{SS}) & 0 & 0 & 0 \\ 0 & 0 & (\lambda, \mu_G^{SS}) & 0 & 0 \\ 0 & 0 & \mu_Q^{SS} & (\lambda, \mu_Q^{SS}) & 0 \\ 0 & 0 & 0 & 0 & (\lambda, \mu_T^S) \end{bmatrix}, \quad (66)$$

where $(x, y) = -(x + y)$ and $(x, y, z) = -(x + y + z)$. Here, the matrix $\mathbf{B}_{10}$ defines the rate of transitions from states in $\hat{\Omega}_1$ to the state $\hat{\Omega}_0$, the matrix $\mathbf{B}_{01}$ defines the rate of transitions from the state $\hat{\Omega}_0$ to states in $\hat{\Omega}_1$, and the offdiagonal elements of the matrix $\mathbf{B}_{11}$ define the rate of transitions among states in $\hat{\Omega}_1$. Similarly, for the homogeneous portion of the state space $\hat{\Omega}$, the matrix $\mathbf{A}_{-1}$ defines the rate of transitions from states in $\hat{\Omega}_{i+1}$ to states in $\hat{\Omega}_i$, the matrix $\mathbf{A}_1$ defines the rate of transitions from states in $\hat{\Omega}_i$ to states in $\hat{\Omega}_{i+1}$, and the offdiagonal elements of the matrix $\mathbf{A}_0$ define the rate of transitions among states in $\hat{\Omega}_i$, $i \in \mathbb{Z}_{\geq 1}$. Hence, the matrices for the homogeneous portion of the state space are given by

$\mathbf{A}_1 = \lambda \mathbf{I}$, $\mathbf{A}_0 = \mathbf{B}_{11}$, and

$$\mathbf{A}_{-1} = \begin{bmatrix} \mathbf{A}_{-1}^{PP} & \mathbf{A}_{-1}^{PS} \\ \mathbf{A}_{-1}^{SP} & \mathbf{A}_{-1}^{SS} \end{bmatrix}, \quad (22)$$

$$\mathbf{A}_{-1}^{PP} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{\parallel} \mu_T^P & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{-1}^{PS} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (1-p_{\parallel})\mu_T^P & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (67)$$

$$\mathbf{A}_{-1}^{SP} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{\parallel} \mu_T^S & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{-1}^{SS} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (1-p_{\parallel})\mu_T^S & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (68)$$

where $\mathbf{I}$ denotes the identity matrix of appropriate dimension.

C Numerical Experiments

In this appendix we present additional details and empirical results from our numerical experiments in Section 7.